

QUOTIENTS OF A THEORY OF PRESHEAF TYPE

In Chapter 3, we have described the classifying topos of a quotient of a geometric theory in a syntactic way. In this chapter we shall study the quotients of a given theory of presheaf type $\mathbb{T}$ by means of Grothendieck topologies that can be naturally attached to them, thereby establishing a ‘semantic’ representation for the classifying topos of such a quotient as a subtopos of the classifying topos of $\mathbb{T}$.

In section 8.1 we discuss such ‘semantic’ representations and introduce, given a theory of presheaf type $\mathbb{T}$ classified by a topos $[\mathcal{C}, \mathbf{Set}]$ and a Grothendieck topology J on the category $\mathcal{C}^{\text{op}}$, the notion of J -homogeneous $\mathbb{T}$ -model, showing that the models of the quotient of $\mathbb{T}$ corresponding to the subtopos $\mathbf{Sh}(\mathcal{C}^{\text{op}}, J) \hookrightarrow [\mathcal{C}, \mathbf{Set}]$ via Theorem 3.2.5 are precisely the $\mathbb{T}$ -models which are J -homogeneous (if the chosen Morita equivalence for $\mathbb{T}$ is canonical). We obtain explicit axiomatizations for the J -homogeneous models in terms of J and of the formulae which present the finitely presentable $\mathbb{T}$ -models. We then restrict our attention to the classifying toposes of quotients with enough set-based models and establish, for any such quotient, a characterization of the Grothendieck topology corresponding to it in terms of its category of set-based models. We also discuss coherent quotients of a cartesian theory and the Grothendieck topologies which correspond to them, namely the finite-type ones; we show that the lattice operations on Grothendieck topologies naturally restrict to the collection of finite-type ones, and that one can effectively compute the lattice operations on the set of quotients of a given geometric theory by means of calculations in the associated lattices of Grothendieck topologies.

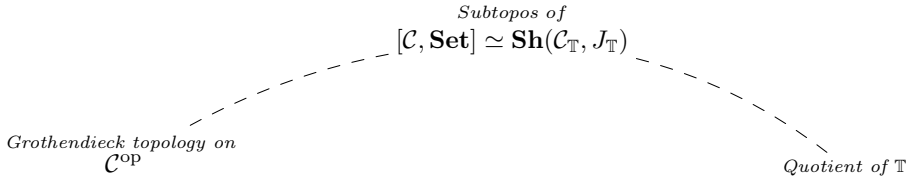
In section 8.2 we identify two main independent conditions which ensure that a quotient of a given theory of presheaf type is again of presheaf type: a finality property of a family of indexed functors naturally associated with the quotient, and a rigidity property of the Grothendieck topology corresponding to the quotient under the above-mentioned duality. We then discuss the problem of finding a geometric theory classified by a given presheaf topos $[\mathcal{K}, \mathbf{Set}]$, and prove a general theorem ensuring that if the category $\mathcal{K}$ can be realized as a full subcategory of the category of finitely presentable models of a theory of presheaf type $\mathbb{T}$, there exists a quotient of $\mathbb{T}$ classified by the topos $[\mathcal{K}, \mathbf{Set}]$, which can be explicitly described in terms of $\mathbb{T}$ and $\mathcal{K}$ provided that some natural conditions are satisfied.

8.1 Studying quotients through the associated Grothendieck topologies

Given a geometric theory $\mathbb{T}$ classified by a presheaf topos via an equivalence

$$[\mathcal{C}, \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}),$$

we have an induced bijective correspondence between the subtoposes of $[\mathcal{C}, \mathbf{Set}]$ and the subtoposes of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, and hence between the Grothendieck topologies on the category $\mathcal{C}^{\text{op}}$ and the quotients of the theory $\mathbb{T}$ (see Theorem 3.2.5):



Under this correspondence, each quotient $\mathbb{T}'$ of $\mathbb{T}$ corresponds to a Grothendieck topology J on $\mathcal{C}^{\text{op}}$ such that the topos $\mathbf{Sh}(\mathcal{C}^{\text{op}}, J)$ classifies $\mathbb{T}'$.

It is natural to wonder if we can characterize the models of $\mathbb{T}'$ among those of $\mathbb{T}$ directly in terms of $\mathbb{T}$ and of J , without any reference to flat functors. As we shall see below, the technique of ‘Yoneda representation of flat functors’ described in section 5.3 provides us with a means for solving this problem.

8.1.1 The notion of J -homogeneous model

This section is based on [17].

By the results of section 6.1.1, we can suppose the Cauchy-completion $\hat{\mathcal{C}}$ of $\mathcal{C}$ to be equal to $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ without loss of generality, and that we have a canonical Morita equivalence

$$\xi^{\mathcal{E}} : \mathbf{Flat}(\hat{\mathcal{C}}^{\text{op}}, \mathcal{E}) \rightarrow \mathbb{T}\text{-mod}(\mathcal{E}).$$

The restriction $\tau_{\xi} : \hat{\mathcal{C}} \rightarrow \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ of the equivalence $\xi^{\mathbf{Set}} : \mathbf{Flat}(\hat{\mathcal{C}}^{\text{op}}, \mathbf{Set}) \simeq \mathbb{T}\text{-mod}(\mathbf{Set})$ is thus isomorphic to the identity, and we have a commutative diagram

$$\begin{array}{ccccc} \mathcal{C} & \xrightarrow{\quad} & \hat{\mathcal{C}} & \xrightarrow[\tau_{\xi}]{\sim} & \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}) \\ y_{\mathcal{C}} \downarrow & & y_{\hat{\mathcal{C}}} \downarrow & & \downarrow i \\ \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathbf{Set}) & \xrightarrow{\sim} & \mathbf{Flat}(\hat{\mathcal{C}}^{\text{op}}, \mathbf{Set}) & \xrightarrow{\xi^{\mathbf{Set}}} & \mathbb{T}\text{-mod}(\mathbf{Set}), \end{array}$$

where $y_{\mathcal{C}} : \mathcal{C} \rightarrow \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathbf{Set})$ and $y_{\hat{\mathcal{C}}} : \hat{\mathcal{C}} \rightarrow \mathbf{Flat}(\hat{\mathcal{C}}^{\text{op}}, \mathbf{Set})$ are the Yoneda embeddings and the horizontal arrows in the left-hand square are the canonical ones.

Recall from Corollary 5.3.2 that every flat functor $F \in \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$ can be represented as

$$F \cong \text{Hom}_{\mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})}^{\mathcal{E}}(\overline{y_{\mathcal{C}}}(-), F),$$

where $\overline{y_{\mathcal{C}}} : \mathcal{C} \rightarrow \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$ is the functor given by the composite

$$\mathcal{C} \xrightarrow{y_{\mathcal{C}}} \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathbf{Set}) \xrightarrow{\gamma_{\mathcal{E}}^* \circ -} \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$$

and $\text{Hom}_{\mathbf{Flat}(\mathcal{C}, \mathcal{E})}^{\mathcal{E}}(\overline{y_{\mathcal{C}}}(-), F)$ is the functor given by the composite

$$\mathcal{C}^{\text{op}} \xrightarrow{\overline{y_{\mathcal{C}}}^{\text{op}} \times \Delta F} \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})^{\text{op}} \times \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E}) \xrightarrow{\text{Hom}_{\mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})}^{\mathcal{E}}} \mathcal{E}.$$

We want to rewrite this Yoneda representation in terms of the theory $\mathbb{T}$.

Recall from section 6.2.1 that the $\mathcal{E}$ -indexed category $\underline{\mathbb{T}\text{-mod}}(\mathcal{E})$ is locally small, with object of morphisms

$$\text{Hom}_{\underline{\mathbb{T}\text{-mod}}(\mathcal{E})}^{\mathcal{E}}(M, N)$$

from M to N in $\mathbb{T}\text{-mod}(\mathcal{E})$. The naturality in $\mathcal{E}$ of the Morita equivalence between $\mathbb{T}$ and the theory of flat functors on $\mathcal{C}^{\text{op}}$ implies the commutativity of the following diagram:

$$\begin{array}{ccc} \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathbf{Set}) & \xrightarrow{\sim} & \mathbb{T}\text{-mod}(\mathbf{Set}) \\ \gamma_{\mathcal{E}}^* \circ - \downarrow & & \downarrow \gamma_{\mathcal{E}}^*(-) \\ \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E}) & \xrightarrow{\sim} & \mathbb{T}\text{-mod}(\mathcal{E}). \end{array}$$

From the commutativity of the two diagrams above we deduce the following representation for F :

$$F \cong \text{Hom}_{\underline{\mathbb{T}\text{-mod}}(\mathcal{E})}^{\mathcal{E}}(\gamma_{\mathcal{E}}^*(i_{\mathcal{C}}(-)), M_F),$$

where M_F is the $\mathbb{T}$ -model in $\mathcal{E}$ corresponding to $F \in \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$ via the above Morita equivalence and $i_{\mathcal{C}} : \mathcal{C} \rightarrow \mathbb{T}\text{-mod}(\mathbf{Set})$ is the canonical induced functor. This motivates the following definition:

Definition 8.1.1 (cf. Definition 2.17 [17]) Let $\mathcal{C}$ be a small category, $\mathcal{E}$ a Grothendieck topos and $\mathbb{T}$ a theory of presheaf type classified by the topos $[\mathcal{C}, \mathbf{Set}]$.

- (a) Given a cosieve S in $\mathcal{C}$ on an object c , a model $M \in \mathbb{T}\text{-mod}(\mathcal{E})$ is said to be S -homogeneous if the family of arrows

$$\text{Hom}_{\underline{\mathbb{T}\text{-mod}}(\mathcal{E})}^{\mathcal{E}}(\gamma_{\mathcal{E}}^*(i_{\mathcal{C}}(d)), M) \xrightarrow{\text{Hom}_{\underline{\mathbb{T}\text{-mod}}(\mathcal{E})}^{\mathcal{E}}(\gamma_{\mathcal{E}}^*(i_{\mathcal{C}}(f)), M)} \text{Hom}_{\underline{\mathbb{T}\text{-mod}}(\mathcal{E})}^{\mathcal{E}}(\gamma_{\mathcal{E}}^*(i_{\mathcal{C}}(c)), M)$$

for $f \in S$, is epimorphic in $\mathcal{E}$.

- (b) Given a Grothendieck topology J on $\mathcal{C}^{\text{op}}$, a model $M \in \mathbb{T}\text{-mod}(\mathcal{E})$ is said to be J -homogeneous if M is S -homogeneous for every $S \in J(c)$.

Remark 8.1.2 It is immediate to see that if J is the atomic topology on $\mathcal{C}^{\text{op}}$ then a model $M \in \mathbb{T}\text{-mod}(\mathcal{E})$ is J -homogeneous if and only if, for every arrow $f : c \rightarrow d$ in $\mathcal{C}$, the arrow

$$\text{Hom}_{\mathbb{T}\text{-mod}(\mathcal{E})}^{\mathcal{E}}(\gamma_{\mathcal{E}}^*(i_{\mathcal{C}}(d)), M) \xrightarrow{\text{Hom}_{\mathbb{T}\text{-mod}(\mathcal{E})}^{\mathcal{E}}(\gamma_{\mathcal{E}}^*(i_{\mathcal{C}}(f)), M)} \text{Hom}_{\mathbb{T}\text{-mod}(\mathcal{E})}^{\mathcal{E}}(\gamma_{\mathcal{E}}^*(i_{\mathcal{C}}(c)), M)$$

is an epimorphism in $\mathcal{E}$. In this case we shall simply say ‘homogeneous’ instead of ‘ J -homogeneous’.

Since the model M_F is J -homogeneous if and only if the functor F is J -continuous, we have the following result:

Theorem 8.1.3 (cf. Theorem 2.19 [17]). *Let $\mathbb{T}$ be a theory of presheaf type classified by a topos $[\mathcal{C}, \mathbf{Set}]$ and J a Grothendieck topology on $\mathcal{C}^{\text{op}}$. Then the subtopos $\mathbf{Sh}(\mathcal{C}^{\text{op}}, J) \hookrightarrow [\mathcal{C}, \mathbf{Set}]$ classifies the J -homogeneous $\mathbb{T}$ -models; that is, for any quotient $\mathbb{T}'$ of $\mathbb{T}$, the $\mathbb{T}'$ -models are exactly the J -homogeneous $\mathbb{T}$ -models in any Grothendieck topos if and only if $\mathbb{T}'$ is classified by the topos $\mathbf{Sh}(\mathcal{C}^{\text{op}}, J)$ and the diagram in $\mathcal{CAT}$*

$$\begin{array}{ccc} \mathbb{T}'\text{-mod}(\mathcal{E}) & \xrightarrow{\cong} & \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}^{\text{op}}, J)) \\ \downarrow i_{\mathbb{T}'}^{\mathcal{E}} & & \downarrow j \circ - \\ \mathbb{T}\text{-mod}(\mathcal{E}) & \xrightarrow[\chi^{\mathcal{E}}]{\cong} & \mathbf{Geom}(\mathcal{E}, [\mathcal{C}, \mathbf{Set}]) \end{array}$$

commutes (up to isomorphism) naturally in $\mathcal{E} \in \mathfrak{BT}\mathbf{op}$, where χ is a canonical Morita equivalence for the theory $\mathbb{T}$,

$$j : \mathbf{Sh}(\mathcal{C}^{\text{op}}, J) \hookrightarrow [\mathcal{C}, \mathbf{Set}]$$

is the canonical geometric inclusion and

$$i_{\mathbb{T}'}^{\mathcal{E}} : \mathbb{T}'\text{-mod}(\mathcal{E}) \hookrightarrow \mathbb{T}\text{-mod}(\mathcal{E})$$

is the inclusion functor. □

Specializing the theorem to the case of the atomic topology gives the following result:

Corollary 8.1.4 (cf. Corollary 2.20 [17]). *Let $(\mathcal{C}^{\text{op}}, J)$ be an atomic site and $\mathbb{T}$ a theory classified by a topos $[\mathcal{C}, \mathbf{Set}]$. Then the subtopos $\mathbf{Sh}(\mathcal{C}^{\text{op}}, J) \hookrightarrow [\mathcal{C}, \mathbf{Set}]$ classifies the homogeneous $\mathbb{T}$ -models.* □

Let us now rephrase in more explicit terms what it means for a model to be J -homogeneous. For this, we first express the condition that a given family of arrows as in Definition 8.1.1 be epimorphic as the validity of a sentence written in the internal language of the topos, and then use Kripke-Joyal semantics to spell out what it means for that sentence to be valid in terms of generalized elements.

Recall that if $\mathcal{E}$ is a Grothendieck topos and $\{f_i : C_i \rightarrow C \mid i \in I\}$ is a family of arrows in it indexed by a set I , then this family is epimorphic if and only if the logical sentence $(\forall y^{\ulcorner C \urcorner})(\bigvee_{i \in I}(\exists x^{\ulcorner C_i \urcorner}(\ulcorner f_i \urcorner(x) = y)))$ written in the internal language $\Sigma_{\mathcal{E}}$ of the topos $\mathcal{E}$ (cf. section 1.3.1.1) holds in the tautological $\Sigma_{\mathcal{E}}$ -structure in $\mathcal{E}$. Given a separating family of objects $\mathcal{S}$ for $\mathcal{E}$, the validity in $\mathcal{E}$ of this sentence is in turn equivalent, by Kripke-Joyal semantics, to the following condition: for any object $E \in \mathcal{S}$ and arrow $y : E \rightarrow C$, there exist an epimorphic family $\{r_i : E_i \rightarrow E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and generalized elements $\{x_i : E_i \rightarrow C_i \mid i \in I\}$ such that $y \circ r_i = f_i \circ x_i$ for each $i \in I$. Applying this to the families of arrows in Definition 8.1.1 and recalling that the objects $\text{Hom}_{\mathbb{T}\text{-mod}(\mathcal{E})}(\gamma_{\mathcal{E}}^*(i_C(d)), M)$ are the objects of morphisms from $\gamma_{\mathcal{E}}^*(i_C(d))$ to M in $\mathbb{T}\text{-mod}(\mathcal{E})$, we obtain the characterization expressed by the following result:

Theorem 8.1.5 (cf. Theorem 2.21 and Corollary 2.22 [17]). *Let $\mathcal{E}$ be a Grothendieck topos with a separating family of objects $\mathcal{S}$ and $\mathbb{T}$ a theory of presheaf type classified by a topos $[\mathcal{C}, \mathbf{Set}]$. Given a Grothendieck topology J on $\mathcal{C}^{\text{op}}$, a model $M \in \mathbb{T}\text{-mod}(\mathcal{E})$ is J -homogeneous if and only if for every cosieve $S \in J(c)$ in $\mathcal{C}$, object $E \in \mathcal{S}$ and arrow $y : !_E^*(\gamma_{\mathcal{E}}^*(i_C(c))) \rightarrow !_E^*(M)$ in $\mathbb{T}\text{-mod}(\mathcal{E}/E)$, there exist an epimorphic family $\{p_f : E_f \rightarrow E, f \in S, E_f \in \mathcal{S}\}$ in $\mathcal{E}$ and for each arrow $f : c \rightarrow d$ in S an arrow $u_f : !_{E_f}^*(\gamma_{\mathcal{E}}^*(i_C(d))) \rightarrow !_{E_f}^*(M)$ in $\mathbb{T}\text{-mod}(\mathcal{E}/E_f)$ such that $p_f^*(y) = u_f \circ !_{E_f}^*(\gamma_{\mathcal{E}}^*(i_C(f)))$.*

In particular, if $\mathcal{E} = \mathbf{Set}$ then a model $M \in \mathbb{T}\text{-mod}(\mathbf{Set})$ is J -homogeneous if and only if for every cosieve $S \in J(c)$ in $\mathcal{C}$ and arrow $y : i_C(c) \rightarrow M$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$, there exist an arrow $f : c \rightarrow d$ in $\mathcal{C}$ belonging to S and an arrow $y : i_C(d) \rightarrow M$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$ such that $y = u_f \circ i_C(f)$. $\square$

By specializing the theorem to the case of the atomic topology, we immediately obtain the following result:

Corollary 8.1.6 (cf. Corollaries 2.23 and 2.24 [17]). *Let $\mathcal{E}$ be a Grothendieck topos with a separating family of objects $\mathcal{S}$ and $\mathbb{T}$ a theory of presheaf type classified by a topos $[\mathcal{C}, \mathbf{Set}]$. If the category $\mathcal{C}^{\text{op}}$ satisfies the right Ore condition then a model $M \in \mathbb{T}\text{-mod}(\mathcal{E})$ is homogeneous if and only if for every arrow $f : c \rightarrow d$ in $\mathcal{C}$, object $E \in \mathcal{S}$ and arrow $y : !_E^*(\gamma_{\mathcal{E}}^*(i_C(c))) \rightarrow !_E^*(M)$ in $\mathbb{T}\text{-mod}(\mathcal{E}/E)$, there exist an object $E_f \in \mathcal{S}$, an epimorphism $p_f : E_f \twoheadrightarrow E$ in $\mathcal{E}$ and an arrow $u_f : !_{E_f}^*(\gamma_{\mathcal{E}}^*(i_C(d))) \rightarrow !_{E_f}^*(M)$ in $\mathbb{T}\text{-mod}(\mathcal{E}/E_f)$ such that $p_f^*(y) = u_f \circ !_{E_f}^*(\gamma_{\mathcal{E}}^*(i_C(f)))$.*

In particular, if $\mathcal{E} = \mathbf{Set}$ then a model $M \in \mathbb{T}\text{-mod}(\mathbf{Set})$ is homogeneous if and only if for any arrows $f : c \rightarrow d$ in $\mathcal{C}$ and $y : i_{\mathcal{C}}(c) \rightarrow M$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$, there exists an arrow $u_f : i_{\mathcal{C}}(d) \rightarrow M$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$ such that $y = u_f \circ i_{\mathcal{C}}(f)$:

$$\begin{array}{ccc} i_{\mathcal{C}}(c) & \xrightarrow{y} & M \\ i_{\mathcal{C}}(f) \downarrow & \nearrow u_f & \\ i_{\mathcal{C}}(d) & & \end{array}$$

□

Remarks 8.1.7 (a) Our notion of homogeneous model essentially specializes to that of (weakly) homogeneous model in classical model theory (cf. Chapter 7 of [49]); in fact, this link has been exploited in [23] to obtain a topos-theoretic interpretation and generalization of Fraïssé's theorem on countably categorical theories (cf. section 10.4 below).

(b) Under the hypotheses of Definition 8.1.1, for any Grothendieck topos $\mathcal{E}$ and any object $E \in \mathcal{E}$, there is an isomorphism

$$!_E^*(\text{Hom}_{\mathbb{T}\text{-mod}(\mathcal{E})}^{\mathcal{E}}(\gamma_{\mathcal{E}}^*(i_{\mathcal{C}}(c)), M)) \cong \text{Hom}_{\mathbb{T}\text{-mod}(\mathcal{E}/E)}^{\mathcal{E}/E}(\gamma_{\mathcal{E}/E}^*(i_{\mathcal{C}}(c)), !_E^*(M))$$

natural in $c \in \mathcal{C}$. Hence, if $M \in \mathbb{T}\text{-mod}(\mathcal{E})$ is J -homogeneous then $!_E^*(M) \in \mathbb{T}\text{-mod}(\mathcal{E}/E)$ is also J -homogeneous. This implies that, when dealing with theories $\mathbb{T}'$ that one wants to prove to satisfy the conditions of Theorem 8.1.5, it suffices to deal with generalized elements defined on $1_{\mathcal{E}}$ by the localization principle. This is illustrated by the example in section 8.1.1.1.

8.1.1.1 Dense linear orders without endpoints as homogeneous models

In this section, we recover, as an application of Corollaries 8.1.4 and 8.1.6, the well-known representation of the classifying topos of the theory of dense linearly ordered objects without endpoints as the category $\mathbf{Sh}(\mathbf{Ord}_{\text{fm}}^{\text{op}}, J_{\text{at}})$ of atomic sheaves on the opposite of the category $\mathbf{Ord}_{\text{fm}}$ of finite ordinals and order-preserving injections between them (cf. Example D3.4.11 [55]).

The theory $\mathbb{L}'$ of dense linearly ordered objects without endpoints is defined over a one-sorted signature consisting of one relation symbol $<$ apart from equality, and has the following axioms:

$$\begin{aligned} & ((x < y) \wedge (y < x) \vdash_{x,y} \perp); \\ & (\top \vdash_{x,y} (x = y) \vee (x < y) \vee (y < x)); \\ & ((x < y) \wedge (y < z) \vdash_{x,y,z} x < z); \\ & (\top \vdash_{\square} (\exists x) \top); \\ & ((x < y) \vdash_{x,y} (\exists z)((x < z) \wedge (z < y))); \end{aligned}$$

$$(x \vdash_x (\exists y, z)((y < x) \wedge (x < z))).$$

The first three axioms give the theory $\mathbb{L}$ of (decidably) linearly ordered objects. It is well known that this theory is of presheaf type (cf. for instance section 9.1 below); the category of finitely presentable $\mathbb{L}$ -models can be identified with $\mathbf{Ord}_{\text{fm}}$, so the classifying topos of $\mathbb{L}$ is equivalent to the presheaf topos $[\mathbf{Ord}_{\text{fm}}, \mathbf{Set}]$. Clearly, the category $\mathbf{Ord}_{\text{fm}}^{\text{op}}$ satisfies the right Ore condition, so we can equip it with the atomic topology J_{at} .

We shall now proceed to verify that the models of $\mathbb{L}'$ are precisely the homogeneous $\mathbb{L}$ -models in every Grothendieck topos $\mathcal{E}$.

A model $M \in \mathbb{L}\text{-mod}(\mathcal{E})$ consists of a pair (I, R) where I is an object of $\mathcal{E}$ and R is a relation on I satisfying the diagrammatic forms of the three axioms of $\mathbb{L}$. We will prove that for each Grothendieck topos $\mathcal{E}$, a model $M = (I, R) \in \mathbb{L}\text{-mod}(\mathcal{E})$ is homogeneous if and only if it is a model of $\mathbb{L}'$, that is if and only if (I, R) is non-empty, dense and without endpoints.

In one direction, let us suppose that M is homogeneous and prove that (I, R) is dense. For each object $E \in \mathcal{E}$, we denote by $<_E$ the order induced by R on $\text{Hom}_{\mathcal{E}}(E, I)$. By the localization principle (cf. Remark 8.1.7(b)), it is enough to prove that if $x, y : 1 \rightarrow I$ are two generalized elements of I with $x <_1 y$ then there exist an object $E \in \mathcal{E}$, an epimorphism $p : E \twoheadrightarrow 1$ and an arrow $z : E \rightarrow I$ such that $x \circ p <_E z <_E y \circ p$. Consider the arrow $f : 2 = \{0, 1\} \rightarrow 3 = \{0, 1, 2\}$ in $\mathbf{Ord}_{\text{fm}}$ defined by $f(0) = 0$ and $f(1) = 2$; the arrows x and y induce, via the assignment $(0 \rightarrow x, 1 \rightarrow y)$ and the universal property of the coproduct $\gamma_{\mathcal{E}}^*(2)$, an arrow $s : \gamma_{\mathcal{E}}^*(2) \rightarrow I$ in $\mathbb{L}\text{-mod}(\mathcal{E})$. The homogeneity of M thus ensures the existence of an object $E \in \mathcal{E}$, an epimorphism $p : E \twoheadrightarrow 1$ and an arrow $t : !_E^*(\gamma_{\mathcal{E}}^*(3)) \rightarrow !_E^*(I)$ in $\mathbb{L}\text{-mod}(\mathcal{E}/E)$ such that $t \circ !_E^*(\gamma_{\mathcal{E}}^*(f)) = !_E^*(s)$. Then the composite arrow

$$E \cong !_E^*(\gamma_{\mathcal{E}}^*(1)) \xrightarrow{!_E^*(\gamma_{\mathcal{E}}^*(u))} !_E^*(\gamma_{\mathcal{E}}^*(3)) \xrightarrow{t} !_E^*(I) \xrightarrow{\pi_I} I,$$

where $u : 1 \rightarrow 3$ is the arrow in $\mathbf{Ord}_{\text{fm}}$ which picks out the element $1 \in 3 = \{0, 1, 2\}$, gives an arrow $z : E \rightarrow I$ with the required properties. The proofs that (I, R) is non-empty and without endpoints are similar and left to the reader.

Conversely, let us prove that if $M \in \mathbb{L}'\text{-mod}(\mathcal{E})$ then M is homogeneous. Again, by the localization principle, this amounts to showing that, for any arrows $f : n \rightarrow m$ in $\mathbf{Ord}_{\text{fm}}$ and $s : \gamma_{\mathcal{E}}^*(n) \rightarrow I$ in $\mathbb{L}\text{-mod}(\mathcal{E})$, there exist an object $E \in \mathcal{E}$, an epimorphism $p : E \twoheadrightarrow 1$ in $\mathcal{E}$ and an arrow $t : !_E^*(\gamma_{\mathcal{E}}^*(m)) \rightarrow !_E^*(I)$ in $\mathbb{L}\text{-mod}(\mathcal{E}/E)$ such that $t \circ !_E^*(\gamma_{\mathcal{E}}^*(f)) = !_E^*(s)$. The arrow s can be identified, via the universal property of the coproduct $\gamma_{\mathcal{E}}^*(n)$, with a family $\{h_i : 1 \rightarrow I \mid i \in n\}$ of generalized elements of I . To find an arrow t satisfying the required property, we inductively use the fourth or fifth axioms of the theory $\mathbb{L}'$ to obtain, starting from the arrows h_i , an object $E \in \mathcal{E}$, an epimorphism $p : E \twoheadrightarrow 1$ in $\mathcal{E}$ and m generalized elements $z_j : E \rightarrow I$ (for $j \in m$) such that, for any $i \in n$, $z_{f(i)} = h_i \circ p$ and, for any $j, j' \in m$, $(j < j')$ implies $(z_j <_E z_{j'})$. The family $\{z_j : E \rightarrow I \mid j \in m\}$ thus yields an arrow $t : !_E^*(\gamma_{\mathcal{E}}^*(m)) \rightarrow !_E^*(I)$ in $\mathbb{L}\text{-mod}(\mathcal{E}/E)$ with the required property.

8.1.2 Axiomatizations for the J -homogeneous models

Given a quotient $\mathbb{T}'$ of a theory of presheaf type $\mathbb{T}$, the Grothendieck topology J on $\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}$ such that the subtopos

$$\mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}, J) \hookrightarrow [\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}]$$

corresponds via the canonical equivalence

$$[\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

to the subtopos of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ associated with $\mathbb{T}'$ via Theorem 3.2.5, will be called the *induced $\mathbb{T}$ -topology of $\mathbb{T}'$* ; note that, by Remark 3.2.6, the canonical Morita equivalence

$$\mathbf{T-mod}(\mathcal{E}) \simeq \mathbf{Flat}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}, \mathcal{E})$$

restricts to a Morita equivalence

$$\mathbb{T}'\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_J(\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}, \mathcal{E}).$$

The following theorem shows that the J -homogeneous models can be axiomatized by geometric sequents over the signature of $\mathbb{T}$.

Theorem 8.1.8 *Let $\mathbb{T}$ be a theory of presheaf type and J a Grothendieck topology on $\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}$. Then there exists a (unique up to syntactic equivalence) quotient $\mathbb{T}'$ of $\mathbb{T}$ such that the $\mathbb{T}'$ -models are exactly the J -homogeneous $\mathbb{T}$ -models in every Grothendieck topos.*

Proof Via the canonical equivalence $[\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ of classifying toposes for $\mathbb{T}$, the geometric inclusion

$$\mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}, J) \hookrightarrow [\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}]$$

corresponds to a subtopos of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$; the quotient of $\mathbb{T}$ corresponding to this inclusion via the duality of Theorem 3.2.5 thus axiomatizes the J -homogeneous $\mathbb{T}$ -models by Remark 3.2.6 and the discussion preceding Theorem 8.1.8. $\square$

Remark 8.1.9 If the category $\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}$ satisfies the right Ore condition and J_{at} is the atomic topology on it, then the geometric inclusion

$$\mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}, J_{\mathrm{at}}) \hookrightarrow [\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}]$$

corresponds to the subtopos $\mathbf{sh}_{\neg\neg}(\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}))$ of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, and hence the J_{at} -homogeneous models are axiomatized by the Booleanization of $\mathbb{T}$ (cf. section 4.2.3). Analogously, the quotient of $\mathbb{T}$ corresponding to the De Morgan topology on the category $\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}$ (cf. section 2 of [13] for the definition of this topology) is precisely the DeMorganization of $\mathbb{T}$ (as defined in section 4.2.3).

The following results provide, for any theory of presheaf type $\mathbb{T}$, explicit descriptions of the induced $\mathbb{T}$ -topology of a given quotient of $\mathbb{T}$ and, reciprocally, of the quotient of $\mathbb{T}$ corresponding to a given Grothendieck topology on the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$.

Given a sequent σ of the form $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} (\exists \vec{y}_i) \theta_i)$, where, for any $i \in I$, $[\theta_i] : \{\vec{y}_i \cdot \psi\} \rightarrow \{\vec{x} \cdot \phi\}$ is an arrow in $\mathcal{C}_{\mathbb{T}}$ and $\phi(\vec{x})$, $\psi(\vec{y}_i)$ are formulae presenting respectively $\mathbb{T}$ -models $M_{\{\vec{x} \cdot \phi\}}$ and $M_{\{\vec{y}_i \cdot \psi_i\}}$, we shall denote by S_{σ} the sieve on the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ defined as follows. For each $i \in I$, $[[\vec{y}_i, \vec{x} \cdot \theta_i]]_{M_{\{\vec{y}_i \cdot \psi_i\}}}$ is the graph of a map $[[\vec{y}_i \cdot \psi_i]]_{M_{\{\vec{y}_i \cdot \psi_i\}}} \rightarrow [[\vec{x} \cdot \phi]]_{M_{\{\vec{y}_i \cdot \psi_i\}}}$; the image under it of the generators of $M_{\{\vec{y}_i \cdot \psi_i\}}$ is an element of $[[\vec{x} \cdot \phi]]_{M_{\{\vec{y}_i \cdot \psi_i\}}}$ and this in turn determines, by definition of $M_{\{\vec{x} \cdot \phi\}}$, a unique arrow $s_i : M_{\{\vec{x} \cdot \phi\}} \rightarrow M_{\{\vec{y}_i \cdot \psi_i\}}$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$. We define S_{σ} to be the sieve in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ on $M_{\{\vec{x} \cdot \phi\}}$ generated by the arrows s_i as i varies in I . Notice that S_{σ} is the sieve on $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ corresponding under the equivalence $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}} \simeq \mathcal{C}_{\mathbb{T}}^{\text{irr}}$ of Theorem 6.1.13 to the sieve in $\mathcal{C}_{\mathbb{T}}^{\text{irr}}$ generated by the arrows $[\theta_i] : \{\vec{y}_i \cdot \psi\} \rightarrow \{\vec{x} \cdot \phi\}$ (for $i \in I$); indeed, for each $i \in I$, the arrow s_i defined above coincides with the arrow $M_{[\theta_i]}$ defined in the proof of Theorem 6.1.13.

Theorem 8.1.10 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ and $\mathbb{T}'$ a quotient of $\mathbb{T}$ obtained from $\mathbb{T}$ by adding axioms σ of the form $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} (\exists \vec{y}_i) \theta_i)$ where, for each $i \in I$, $[\theta_i] : \{\vec{y}_i \cdot \psi\} \rightarrow \{\vec{x} \cdot \phi\}$ is an arrow in $\mathcal{C}_{\mathbb{T}}$ and $\phi(\vec{x})$, $\psi(\vec{y}_i)$ are geometric formulae over Σ presenting respectively $\mathbb{T}$ -models $M_{\{\vec{x} \cdot \phi\}}$ and $M_{\{\vec{y}_i \cdot \psi_i\}}$ (note that, by Theorem 3.2.5, every quotient of $\mathbb{T}$ has an axiomatization of this form, cf. also Theorem 8.1.12 below). With the above notation, the induced $\mathbb{T}$ -topology of $\mathbb{T}'$ is generated by the sieves S_{σ} .*

Proof Let $F : \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}} \rightarrow \mathcal{E}$ be a flat functor; if $M_{\{\vec{x} \cdot \phi\}}$ is a finitely presented $\mathbb{T}$ -model then $F(M_{\{\vec{x} \cdot \phi\}}) = [[\vec{x} \cdot \phi]]_{M_F}$ where M_F is the $\mathbb{T}$ -model in $\mathcal{E}$ corresponding to F via the canonical Morita equivalence for $\mathbb{T}$. Indeed, denoting by $g : \mathcal{E} \rightarrow \mathbf{Set}[\mathbb{T}] \simeq [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ the geometric morphism corresponding to F via the universal property of the classifying topos of $\mathbb{T}$, we have that $F = g^* \circ y$ where $y : \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}} \rightarrow [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ is the Yoneda embedding. But $M_F = g^*(N_{\mathbb{T}})$ where $N_{\mathbb{T}}$ is the universal model of $\mathbb{T}$ lying in the classifying topos $\mathbf{Set}[\mathbb{T}] \simeq [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ (as in Theorem 6.1.1), and the representable $\text{Hom}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}(M_{\{\vec{x} \cdot \phi\}}, -) \in [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ is clearly isomorphic to $[[\vec{x} \cdot \phi]]_{N_{\mathbb{T}}}$. Since inverse image functors of geometric morphisms preserve the interpretations of geometric formulae, we may conclude that $F(M_{\{\vec{x} \cdot \phi\}}) = [[\vec{x} \cdot \phi]]_{M_F}$, as required. It is also immediate to see that, for any arrow $[\theta_i] : \{\vec{y}_i \cdot \psi\} \rightarrow \{\vec{x} \cdot \phi\}$ in $\mathcal{C}_{\mathbb{T}}$, the graph of the arrow $F(M_{[\theta_i]})$ is precisely $[[\vec{y}_i, \vec{x} \cdot \theta_i]]_{M_F}$.

Now, for any sequent σ as in the statement of the theorem, a model $N \in \mathbb{T}\text{-mod}(\mathcal{E})$ is S_{σ} -homogeneous if and only if σ holds in N ; indeed, N is S_{σ} -homogeneous if and only if F_N sends S_{σ} to an epimorphic family, that is, by the

above discussion, if and only if σ holds in N . So, if J is the induced $\mathbb{T}$ -topology of $\mathbb{T}'$, from the equivalence

$$\mathbb{T}'\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_J(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E})$$

it follows that for any $\mathcal{E} \in \mathfrak{BTOP}$ and any $F \in \mathbf{Flat}_J(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E})$, F sends S_σ to an epimorphic family for any axiom σ of $\mathbb{T}'$ as in the statement of the theorem. This implies, by Lemma 3 at p. 393 of [63] applied to the canonical geometric inclusion

$$\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J) \hookrightarrow [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}],$$

that for any such σ the associated sheaf functor

$$a_J : [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \rightarrow \mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J)$$

sends the monomorphism $S_\sigma \mapsto \text{Hom}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}(M_{\{\vec{x}.\phi\}}, -)$ to an isomorphism i.e. that S_σ is J -covering. Thus the Grothendieck topology J' generated by the sieves S_σ is contained in J . To prove that $J' = J$, it is equivalent to verify, by Diaconescu's equivalence and the 2-dimensional Yoneda lemma (cf. section 3.1.1), that for any Grothendieck topos $\mathcal{E}$, $\mathbf{Flat}_{J'}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E}) = \mathbf{Flat}_J(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E})$. Now, since $J' \subseteq J$, $\mathbf{Flat}_J(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E}) \subseteq \mathbf{Flat}_{J'}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E})$. To prove the other inclusion, observe that, since J' contains all the sieves S_σ , the $\mathbb{T}$ -model corresponding to a functor $F \in \mathbf{Flat}_{J'}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E})$ belongs to $\mathbb{T}'\text{-mod}(\mathcal{E})$ and hence, by the equivalence $\mathbb{T}'\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_J(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E})$, F is J -continuous, as required. $\square$

Remarks 8.1.11 (a) Theorem 8.1.10 generalizes the method of construction of the classifying topos of a quotient of a cartesian theory given by Propositions D3.1.7 and D3.1.10 [55] (cf. also the theorem at p. 25 of [39] for the particular case of a Horn theory); indeed, it is well known that the opposite of the category of finitely presentable models of a cartesian theory is equivalent to its cartesian syntactic category (cf. Theorem 2.1.21).

(b) By Theorem 6.1.13, every sieve on $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ is of the form S_σ for some geometric sequent σ as in the statement of Theorem 8.1.10.

(c) Given a quotient $\mathbb{T}'$ of a theory of presheaf type $\mathbb{T}$, one can obtain an axiomatization of $\mathbb{T}'$ of the form required in Theorem 8.1.10 by replacing any axiom $(\phi \vdash_{\vec{x}} \psi)$ of $\mathbb{T}'$, where $\phi(\vec{x})$ and $\psi(\vec{x})$ are geometric formulae over the signature of $\mathbb{T}$, with the collection of sequents corresponding to the sieves S on the category $\mathcal{C}_{\mathbb{T}}^{\text{irr}}$ obtained as follows. By Corollary 6.1.10, there is a $J_{\mathbb{T}}$ -covering sieve T on $\{\vec{x}.\phi\}$ in $\mathcal{C}_{\mathbb{T}}$ generated by arrows f whose domain is a $\mathbb{T}$ -irreducible formula. For each such arrow f , consider the object A_f of $\mathcal{C}_{\mathbb{T}}$ given by the pullback in $\mathcal{C}_{\mathbb{T}}$ of f along the canonical monomorphism $\{\vec{x}.\phi \wedge \psi\} \mapsto \{\vec{x}.\phi\}$; then, again by Corollary 6.1.10, there exists a $J_{\mathbb{T}}$ -covering sieve on A_f generated by arrows k^f whose domains are $\mathbb{T}$ -irreducible geometric formulae.

The sieves S in $\mathcal{C}_{\mathbb{T}}^{\text{irr}}$ on the objects $\text{dom}(f)$ generated by the composites of the arrows k^f with the canonical arrow $A_f \rightarrow \text{dom}(f)$ satisfy the required condition since they are all covering for a Grothendieck topology on $\mathcal{C}_{\mathbb{T}}$ containing $J_{\mathbb{T}}$ if and only if the sieve generated by the canonical monomorphism $\{\vec{x} \cdot \phi \wedge \psi\} \rightarrow \{\vec{x} \cdot \phi\}$ is.

Theorem 8.1.12 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ and J a Grothendieck topology on the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$. Then the unique (up to syntactic equivalence) quotient $\mathbb{T}'$ of $\mathbb{T}$ whose induced $\mathbb{T}$ -topology is J (and hence whose models in every Grothendieck topos are exactly the J -homogeneous $\mathbb{T}$ -models) is axiomatized by the sequents σ over Σ of the form $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} (\exists \vec{y}_i) \theta_i)$ (where, for each $i \in I$, $[\theta_i] : \{\vec{y}_i \cdot \psi\} \rightarrow \{\vec{x} \cdot \phi\}$ is an arrow in $\mathcal{C}_{\mathbb{T}}$ and $\phi(\vec{x})$, $\psi(\vec{y}_i)$ are geometric formulae over Σ which present $\mathbb{T}$ -models) such that the sieve S_{σ} is J -covering. In fact, for any sequent σ of the above form, σ is provable in $\mathbb{T}'$ if and only if S_{σ} is J -covering.*

Proof By Theorem 8.1.3, the quotient $\mathbb{T}'$ of $\mathbb{T}$ whose induced $\mathbb{T}$ -topology is J is characterized by the property that its models in every Grothendieck topos are the J -homogeneous $\mathbb{T}$ -models. From the proof of Theorem 8.1.10 and Remark 8.1.11(b) it thus follows that $\mathbb{T}'$ is axiomatized by the sequents σ whose associated sieves S_{σ} are J -covering.

It thus remains to prove that, for any sequent σ of the form specified above, σ is provable in $\mathbb{T}'$ if and only if S_{σ} is J -covering. Since universal models are preserved by inverse image functors of geometric inclusions (cf. Lemma 2.1 [19]), this follows from the fact that the canonical monomorphism $S_{\sigma} \rightarrow \text{Hom}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}(M_{\{\vec{x} \cdot \phi\}}, -)$ is isomorphic to $[[\vec{x} \cdot \bigvee_{i \in I} (\exists \vec{y}_i) \theta_i]]_{N_{\mathbb{T}}} \rightarrow [[\vec{x} \cdot \phi]]_{N_{\mathbb{T}}}$, where $N_{\mathbb{T}}$ is the universal model of $\mathbb{T}$ defined in Theorem 6.1.1 (cf. Theorem 3.3 [19]), in light of the conservativity of universal models. $\square$

Remark 8.1.13 Theorem 8.1.12, combined with the duality of Theorem 3.2.5, yields a proof-theoretic equivalence between the classical deduction system for geometric logic over $\mathbb{T}$ and the proof system $\mathcal{T}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}}^T$ (where T is the trivial Grothendieck topology) introduced in section 3.2.2 (cf. also Remarks 8.1.11(b)-(c)). This generalizes Coste-Lombardi-Roy's correspondence [40] between dynamical theories (viewed as coherent quotients of a Horn theory) and the finite-type Grothendieck topologies associated with them.

Corollary 8.1.14 *Let $\mathbb{T}$ be a theory of presheaf type such that $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ satisfies the right Ore condition. Then the quotient $\mathbb{T}'$ of $\mathbb{T}$ whose induced $\mathbb{T}$ -topology is the atomic topology on $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ is the theory obtained from $\mathbb{T}$ by adding all the axioms of the form $(\phi \vdash_{\vec{x}} (\exists \vec{y}) \theta)$, where $[\theta] : \{\vec{y} \cdot \psi\} \rightarrow \{\vec{x} \cdot \phi\}$ is any arrow in $\mathcal{C}_{\mathbb{T}}$ and $\phi(\vec{x})$, $\psi(\vec{y})$ are geometric formulae over the signature of $\mathbb{T}$ which present $\mathbb{T}$ -models.*

Proof The thesis follows from Theorem 8.1.12 in light of Remark 8.1.11(b), by observing that the collection of principal sieves on $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ generates the atomic topology on $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$. $\square$

Remark 8.1.15 If $\mathbb{T}$ is cartesian then the hypotheses of Corollary 8.1.14 are satisfied; indeed, the equivalence $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}} \simeq \mathcal{C}_{\mathbb{T}}^{\text{cart}}$ of Theorem 2.1.21 shows that the category $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ is cartesian and hence *a fortiori* satisfies the right Ore condition.

8.1.3 Quotients with enough set-based models

In this section we focus on quotients of theories of presheaf type having enough set-based models. We will extend some ideas and results from sections 3.2 and 3.4 of [68], rewriting them in a general topos-theoretic context. This will lead in particular to a representation for the classifying topos of such a quotient in terms of its category of set-based models, holding under the hypothesis that it has enough points.

Let us first review some basic facts about ind-completions which will be useful for our analysis; we refer the reader to section C4.2 of [55] for more details.

Recall that a finitely accessible category $\mathcal{C}$ is a category which is equivalent to the ind-completion $\text{Ind-}\mathcal{C}$ of a small category $\mathcal{C}$; $\text{Ind-}\mathcal{C}$ is defined to be the full subcategory of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ on the flat functors $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$. Every representable functor is flat, so the Yoneda embedding $y : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ factors through the embedding $\text{Ind-}\mathcal{C} \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$; we will denote this factorization by $y_{\mathcal{C}} : \mathcal{C} \rightarrow \text{Ind-}\mathcal{C}$. Moreover, the inclusion $\text{Ind-}\mathcal{C} \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ creates filtered colimits.

Given a finitely accessible category $\mathcal{L}$, we write $\mathbf{f.p.}\mathcal{L}$ for the full subcategory of $\mathcal{L}$ on the finitely presentable objects; then the embedding $\mathbf{f.p.}\mathcal{L} \hookrightarrow \mathcal{L}$ corresponds to the embedding $y_{\mathbf{f.p.}\mathcal{L}} : \mathbf{f.p.}\mathcal{L} \rightarrow \text{Ind-}\mathcal{L}$ under the canonical equivalence $\mathcal{L} \simeq \text{Ind-}\mathcal{L}$ (cf. Proposition C4.2.2 and Corollary A1.1.9 [55]).

The ind-completion $\text{Ind-}\mathcal{C}$ of a small category $\mathcal{C}$ is the free filtered-colimit completion of $\mathcal{C}$ (cf. Corollary C4.2.6 [55]); that is, for any category $\mathcal{D}$ with filtered colimits, any functor $H : \mathcal{C} \rightarrow \mathcal{D}$ extends, via $y_{\mathcal{C}} : \mathcal{C} \rightarrow \text{Ind-}\mathcal{C}$, uniquely up to canonical isomorphism, to a filtered-colimit-preserving functor $\overline{H} : \text{Ind-}\mathcal{C} \rightarrow \mathcal{D}$. Notice that if $H = \text{Hom}_{\mathcal{C}}(c, -)$ then $\overline{H}$ is naturally isomorphic to $\text{Hom}_{\text{Ind-}\mathcal{C}}(y_{\mathcal{C}}(c), -)$.

Generalizing from section 3.2 of [67], given a small category $\mathcal{C}$, we shall construct correspondences between the collection $\mathcal{S}_{\text{Ind-}\mathcal{C}}$ of full subcategories of $\text{Ind-}\mathcal{C}$ and the collection $\mathcal{G}(\mathcal{C})$ of Grothendieck topologies on $\mathcal{C}^{\text{op}}$. Given a sieve S in $\mathcal{C}^{\text{op}}$ on an object $c \in \mathcal{C}$, we denote by $\overline{S}$ the extension $\text{Ind-}\mathcal{C} \rightarrow \mathbf{Set}$ of $S : \mathcal{C} \rightarrow \mathbf{Set}$ (regarded as a subfunctor of the representable $\text{Hom}_{\mathcal{C}}(c, -)$) along $y_{\mathcal{C}}$ as above.

Let us define two maps $\mathcal{K} : \mathcal{G}(\mathcal{C}) \rightarrow \mathcal{S}_{\text{Ind-}\mathcal{C}}$ and $\mathcal{H} : \mathcal{S}_{\text{Ind-}\mathcal{C}} \rightarrow \mathcal{G}(\mathcal{C})$ as follows.

Given a Grothendieck topology J on $\mathcal{C}^{\text{op}}$, $\mathcal{K}(J)$ is the full subcategory of $\text{Ind-}\mathcal{C}$ defined by

$$d \in \mathcal{K}(J) \text{ iff } \overline{S}(d) = \text{Hom}_{\text{Ind-}\mathcal{C}}(y_{\mathcal{C}}(c), d) \text{ for all } S \in J(c),$$

for any $d \in \text{Ind-}\mathcal{C}$. Here by the equality

$$\overline{S}(d) = \text{Hom}_{\text{Ind-}\mathcal{C}}(y_{\mathcal{C}}(c), d)$$

we mean that the canonical map $\overline{S}(d) \rightarrow \text{Hom}_{\text{Ind-}\mathcal{C}}(y_{\mathcal{C}}(c), d)$ is an isomorphism, equivalently that whenever $d = \text{colim}(y_{\mathcal{C}} \circ G)$ in $\text{Ind-}\mathcal{C}$ for a functor $G : \mathcal{I} \rightarrow \mathcal{C}$ defined on a small filtered category $\mathcal{I}$, for any arrow $r : c \rightarrow G(i)$ there exist objects $j, k \in \mathcal{I}$ and arrows $s : c \rightarrow G(j)$ in S and $\chi : i \rightarrow k$, $\xi : j \rightarrow k$ in $\mathcal{I}$ such that $G(\chi) \circ r = G(\xi) \circ s$.

In the converse direction, given a full subcategory $\mathcal{D}$ of $\text{Ind-}\mathcal{C}$, we define $\mathcal{H}(\mathcal{D})$ by

$$S \in \mathcal{H}(\mathcal{D})(c) \text{ iff } \overline{S}(d) = \text{Hom}_{\text{Ind-}\mathcal{C}}(y_{\mathcal{C}}(c), d) \text{ for all } d \in \mathcal{D},$$

for any sieve S in $\mathcal{C}^{\text{op}}$ on an object c .

It is easy to verify that for any full subcategory $\mathcal{D}$ of $\text{Ind-}\mathcal{C}$, $\mathcal{H}(\mathcal{D})$ is indeed a Grothendieck topology on $\mathcal{C}^{\text{op}}$; we provide the details for the reader's convenience. It is clear that the maximality axiom holds. Let us verify the stability axiom. Given an arrow $f : c \rightarrow c'$ in $\mathcal{C}$ and a sieve $S \in \mathcal{H}(\mathcal{D})(c)$, we want to prove that $f^*(S) \in \mathcal{H}(\mathcal{D})(c')$, i.e. that for any functor $G : \mathcal{I} \rightarrow \mathcal{C}$ defined on a small filtered category $\mathcal{I}$ and any arrow $r : c' \rightarrow G(i)$, there exist $j, k \in \mathcal{I}$ and arrows $s : c' \rightarrow G(j)$ in $f^*(S)$ and $\chi : i \rightarrow k$, $\xi : j \rightarrow k$ in $\mathcal{I}$ such that $G(\chi) \circ r = G(\xi) \circ s$. Consider the arrow $r \circ f$; since $S \in \mathcal{H}(\mathcal{D})(c)$, there exist $j', k' \in \mathcal{I}$ and arrows $s' : c \rightarrow G(j')$ in S and $\chi' : i \rightarrow k'$, $\xi' : j' \rightarrow k'$ in $\mathcal{I}$ such that $G(\chi') \circ r \circ f = G(\xi') \circ s'$. Then if we take $i = i'$, $j = k = k'$, $\chi = \chi'$, $\xi = 1_k$ and $s = G(\chi) \circ r$, we have that $s \in f^*(S)$ and hence our thesis is satisfied. It remains to verify the transitivity axiom. Given a sieve R in $\mathcal{C}^{\text{op}}$ on an object c and a sieve $S \in \mathcal{H}(\mathcal{D})(c)$ such that $f^*(R) \in \mathcal{H}(\mathcal{D})(c')$ for any arrow $f : c \rightarrow c'$ in $\mathcal{C}$ belonging to S , we want to prove that $R \in \mathcal{H}(\mathcal{D})(c)$. Since $S \in \mathcal{H}(\mathcal{D})(c)$, given an arrow $r : c \rightarrow G(i)$ in $\mathcal{C}$ there exist $j, k \in \mathcal{I}$ and arrows $f' : c \rightarrow G(j)$ in S and $\chi : i \rightarrow k$, $\xi : j \rightarrow k$ in $\mathcal{I}$ such that $G(\chi) \circ r = G(\xi) \circ f'$; now, since $f^*(R) \in \mathcal{H}(\mathcal{D})(c')$, there exist $j', k' \in \mathcal{I}$ and arrows $g : G(j) \rightarrow G(j')$ in $f^*(R)$ and $\chi' : k \rightarrow k'$, $\xi' : j' \rightarrow k'$ in $\mathcal{I}$ such that $G(\chi') \circ G(\xi) = G(\xi') \circ g$; so $G(\chi' \circ \chi) \circ r = G(\xi') \circ g \circ f'$ and our thesis is satisfied.

Next, we note that $\mathcal{S}_{\text{Ind-}\mathcal{C}}$ and $\mathcal{G}(\mathcal{C})$ are naturally equipped with partial orders (respectively the inclusion ordering between full subcategories of $\mathcal{S}_{\text{Ind-}\mathcal{C}}$ and the inclusion ordering between Grothendieck topologies on $\mathcal{C}^{\text{op}}$) and if we regard them as poset categories then the correspondences $\mathcal{H}$ and $\mathcal{K}$ become contravariant functors; moreover, it is immediate to see that they form a Galois connection between $\mathcal{S}_{\text{Ind-}\mathcal{C}}$ and $\mathcal{G}(\mathcal{C})$. From the formal theory of Galois connections, it thus follows that $\mathcal{H}(\mathcal{K}(\mathcal{H}(\mathcal{D}))) = \mathcal{H}(\mathcal{D})$ for any full subcategory $\mathcal{D}$ of $\text{Ind-}\mathcal{C}$ and $\mathcal{K}(\mathcal{H}(\mathcal{K}(J))) = \mathcal{K}(J)$ for any Grothendieck topology J on $\mathcal{C}^{\text{op}}$.

The following lemma represents the extension of Lemma 3.11 [68] to the context of finitely accessible categories.

Lemma 8.1.16 *Let J be a Grothendieck topology on a small category $\mathcal{C}^{\text{op}}$. Then, with the above notation, the functor $\text{Hom}_{\text{Ind-}\mathcal{C}}(y_{\mathcal{C}}(-), d) : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ is J -continuous if and only if $d \in \mathcal{K}(J)$, for any $d \in \text{Ind-}\mathcal{C}$.*

Proof Recall from section 1.1.5 that via the equivalence

$$\mathbf{Geom}(\mathbf{Set}, [\mathcal{C}, \mathbf{Set}]) \simeq \mathbf{Flat}(\mathcal{C}^{\mathrm{op}}, \mathbf{Set})$$

a flat functor $F : \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$ corresponds to the geometric morphism having as inverse image $F \otimes_{\mathcal{C}} - \cong - \otimes_{\mathcal{C}^{\mathrm{op}}} F : [\mathcal{C}, \mathbf{Set}] \rightarrow \mathbf{Set}$ (cf. section 1.1.5 for the notation); so F is J -continuous (for a Grothendieck topology J on $\mathcal{C}^{\mathrm{op}}$) if and only if for every $S \in J(c)$, $F \otimes_{\mathcal{C}} -$ sends the monomorphism $S \rightarrow \mathrm{Hom}_{\mathcal{C}}(c, -)$ to an isomorphism. The functor $\mathrm{Hom}_{\mathrm{Ind}\text{-}\mathcal{C}}(y_{\mathcal{C}}(-), d) : \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$ is flat for any $d \in \mathrm{Ind}\text{-}\mathcal{C}$ (by the Yoneda representation of flat functors, cf. section 5.3), so it is J -continuous if and only if for every $S \in J(c)$, $\mathrm{Hom}_{\mathrm{Ind}\text{-}\mathcal{C}}(y_{\mathcal{C}}(-), d) \otimes_{\mathcal{C}} -$ sends $S \rightarrow \mathrm{Hom}_{\mathcal{C}}(c, -)$ to an isomorphism.

Now, if $d = \mathrm{colim}(y_{\mathcal{C}} \circ G)$ in $\mathrm{Ind}\text{-}\mathcal{C}$ for a functor $G : \mathcal{I} \rightarrow \mathcal{C}$ defined on a small filtered category $\mathcal{I}$ then

$$\mathrm{Hom}_{\mathrm{Ind}\text{-}\mathcal{C}}(y_{\mathcal{C}}(-), d) \cong \mathrm{colim}_{[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]} \mathrm{Hom}_{\mathrm{Ind}\text{-}\mathcal{C}}(y_{\mathcal{C}}(-), G(-)),$$

since all the objects of $\mathcal{C}$ are finitely presentable in $\mathrm{Ind}\text{-}\mathcal{C}$ and colimits in functor categories are computed pointwise. So, for any functor $P : \mathcal{C} \rightarrow \mathbf{Set}$, since $(- \otimes_{\mathcal{C}} P)$ preserves filtered colimits (having a right adjoint) and $\mathrm{Hom}_{\mathcal{C}}(-, c) \otimes_{\mathcal{C}} P \cong P(c)$ for any $c \in \mathcal{C}$ (cf. the proof of Theorem 1.1.31), we have that $\mathrm{Hom}_{\mathrm{Ind}\text{-}\mathcal{C}}(y_{\mathcal{C}}(-), d) \otimes_{\mathcal{C}} P \cong \mathrm{colim}(P \circ G) = \bar{P}(d)$. Hence $\mathrm{Hom}_{\mathrm{Ind}\text{-}\mathcal{C}}(y_{\mathcal{C}}(-), d) \otimes_{\mathcal{C}} -$ sends $S \rightarrow \mathrm{Hom}_{\mathcal{C}}(c, -)$ to the monomorphism $\bar{S}(d) \rightarrow \mathrm{Hom}_{\mathrm{Ind}\text{-}\mathcal{C}}(y_{\mathcal{C}}(c), d)$, from which our thesis follows. $\square$

Remark 8.1.17 The proof of Lemma 8.1.16 actually shows that for any small category $\mathcal{C}$, sieve S in $\mathcal{C}^{\mathrm{op}}$ and object d of $\mathrm{Ind}\text{-}\mathcal{C}$, the equality

$$\bar{S}(d) = \mathrm{Hom}_{\mathrm{Ind}\text{-}\mathcal{C}}(y_{\mathcal{C}}(c), d)$$

holds if and only if d (regarded as a set-based model of the theory of flat functors on $\mathcal{C}^{\mathrm{op}}$) is S -homogeneous in the sense of Definition 8.1.1(a). It follows that for any Grothendieck topology J on $\mathcal{C}^{\mathrm{op}}$, $\mathcal{K}(J)$ is the full subcategory of $\mathrm{Ind}\text{-}\mathcal{C}$ on the J -homogeneous objects, and that for any full subcategory $\mathcal{D}$ of $\mathrm{Ind}\text{-}\mathcal{C}$, the $\mathcal{H}(\mathcal{D})$ -covering sieves are precisely the sieves S in $\mathcal{C}^{\mathrm{op}}$ such that every object of $\mathcal{D}$ is S -homogeneous.

Proposition 8.1.18 *Let $\mathbb{T}$ be a theory of presheaf type and $\mathbb{T}'$ a quotient of $\mathbb{T}$. Then, denoting by $\mathbb{T}'\text{-mod}(\mathbf{Set})$ the full subcategory of $\mathbb{T}\text{-mod}(\mathbf{Set})$ on the $\mathbb{T}'$ -models, we have $\mathcal{K}(\mathcal{H}(\mathbb{T}'\text{-mod}(\mathbf{Set}))) = \mathbb{T}'\text{-mod}(\mathbf{Set})$.*

Proof By Theorems 3.2.5 and 8.1.3, there exists a (unique) Grothendieck topology J on $\mathrm{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}$ such that the $\mathbb{T}'$ -models are exactly the J -homogeneous ones in every Grothendieck topos (cf. also the proof of Theorem 8.1.8); but by Lemma 8.1.16 a $\mathbb{T}$ -model M in $\mathbf{Set}$ is J -homogeneous if and only if $M \in \mathcal{K}(J)$, whence $\mathbb{T}'\text{-mod}(\mathbf{Set}) = \mathcal{K}(J)$. Our thesis thus follows from the discussion preceding Lemma 8.1.16. $\square$

Remark 8.1.19 By Theorem 8.1.8 and Remark 8.1.17, every full subcategory of $\mathbb{T}\text{-mod}(\mathbf{Set})$ of the form $\mathcal{K}(J)$ (for a Grothendieck topology J on the category $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$) is of the form $\mathbb{T}'\text{-mod}(\mathbf{Set})$ for a quotient $\mathbb{T}'$ of $\mathbb{T}$. It thus follows from Proposition 8.1.18 and the discussion preceding Lemma 8.1.16 that, in the case of categories $\mathcal{C}$ of the form $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ for a theory of presheaf type $\mathbb{T}$, the ‘closed’ full subcategories of our Galois correspondence are precisely the categories of set-based models of quotients of $\mathbb{T}$.

We are now ready to prove the main result of this section.

Theorem 8.1.20 *Let $\mathbb{T}$ be a theory of presheaf type and $\mathbb{T}'$ be a quotient of $\mathbb{T}$ having enough set-based models. Then the topos*

$$\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{H}(\mathbb{T}'\text{-mod}(\mathbf{Set})))$$

classifies $\mathbb{T}'$ (via Theorem 3.2.5), provided that it has enough points.

Proof From Theorem 8.1.8 we know that there exists a quotient $\mathbb{T}''$ of $\mathbb{T}$ classified by the topos $\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{H}(\mathbb{T}'\text{-mod}(\mathbf{Set})))$ whose models in any Grothendieck topos are exactly the $\mathcal{H}(\mathbb{T}'\text{-mod}(\mathbf{Set}))$ -homogeneous $\mathbb{T}$ -models. Now, if the topos $\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{H}(\mathbb{T}'\text{-mod}(\mathbf{Set})))$ has enough points then $\mathbb{T}''$ has enough set-based models (cf. Proposition 4.2.20). But $\mathbb{T}''$ has the same set-based models as the theory $\mathbb{T}'$ by Lemma 8.1.16 and Proposition 8.1.18 and $\mathbb{T}'$ has enough set-based models by our hypotheses, so we may conclude that $\mathbb{T}'$ and $\mathbb{T}''$ are syntactically equivalent and hence that they have equivalent classifying toposes; in particular, the topos

$$\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{H}(\mathbb{T}'\text{-mod}(\mathbf{Set})))$$

classifies $\mathbb{T}'$ via Theorem 3.2.5, as required. $\square$

From the proof of Theorem 8.1.20, we can extract the following result:

Proposition 8.1.21 *Let $\mathbb{T}$ be a theory of presheaf type and J a Grothendieck topology on the category $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ such that both the toposes*

$$\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J)$$

and

$$\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{H}(\mathcal{K}(J)))$$

have enough points. Then $J = \mathcal{H}(\mathcal{K}(J))$.

Proof By the results of section 1.3.4, $J = \mathcal{H}(\mathcal{K}(J))$ if and only if there exists an equivalence

$$\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J) \simeq \mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{H}(\mathcal{K}(J)))$$

which commutes (in the obvious sense) with the canonical geometric inclusions

$$\mathbf{Sh}(\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, J) \hookrightarrow [\mathbf{f.p.T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

and

$$\mathbf{Sh}(\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, \mathcal{H}(\mathcal{K}(J))) \hookrightarrow [\mathbf{f.p.T}\text{-mod}(\mathbf{Set}), \mathbf{Set}];$$

but this is equivalent, by the 2-dimensional Yoneda lemma (cf. section 3.1.1) and the universal property of classifying toposes, to saying that the quotients $\mathbb{T}'$ and $\mathbb{T}''$ of $\mathbb{T}$ classified respectively by

$$\mathbf{Sh}(\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, J)$$

and

$$\mathbf{Sh}(\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, \mathcal{H}(\mathcal{K}(J)))$$

via Theorem 8.1.8 have exactly the same models in every Grothendieck topos. Now, if both $\mathbb{T}'$ and $\mathbb{T}''$ have enough set-based models then this happens precisely when they have exactly the same set-based models, equivalently (by Lemma 8.1.16) when $\mathcal{K}(J) = \mathcal{K}(\mathcal{H}(\mathcal{K}(J)))$; but this holds by the formal properties of Galois connections (cf. the discussion preceding Lemma 8.1.16). $\square$

Remark 8.1.22 Theorem 8.1.20 and Proposition 8.1.21 can be notably applied in the case of a coherent quotient of a cartesian theory $\mathbb{T}$. Indeed, assuming the axiom of choice, if $\mathbb{T}'$ is a coherent quotient of $\mathbb{T}$ then it readily follows that $\mathcal{H}(\mathbb{T}'\text{-mod}(\mathbf{Set}))$ is a topology of finite type in the sense of Definition 8.1.23 (cf. the proof of Theorem 8.1.10 and Theorem 10.8.6(iii)). Conversely, if J is a topology of finite type on $\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}$ then there exists a (unique up to syntactic equivalence, assuming the axiom of choice) coherent quotient $\mathbb{T}'$ of $\mathbb{T}$ such that $\mathbb{T}'\text{-mod}(\mathbf{Set}) = \mathcal{K}(J)$. See also Proposition 8.1.27 below.

Finally, let us discuss how our results relate to those in sections 3.2 and 3.4 of [68]. There the authors dealt with the case of embeddings $\mathcal{C} \hookrightarrow \mathrm{Ind}\text{-}\mathcal{C}$ of the form $\mathbf{f.p.}\mathcal{L} \hookrightarrow \mathcal{L}$ for a locally finitely presentable category $\mathcal{L}$. It is well known that these categories $\mathcal{L}$ are precisely the categories of set-based models of cartesian theories and that for any such category $\mathcal{L}$ the category $\mathbf{f.p.}\mathcal{L}$ admits a syntactic description as the opposite of the syntactic category of the relevant cartesian theory (cf. Theorem 2.1.21). This fact is exploited in an essential way to derive some results in [68], for example Proposition 3.5. Instead, we have arrived at Proposition 8.1.18, which generalizes Proposition 3.5, by using the theory of classifying toposes and Theorem 3.2.5. Also, our Lemma 8.1.16 generalizes Lemma 3.11 [68], whose proof relied on the locally finite presentability of the category $\mathcal{L}$, and our Proposition 8.1.21 implies Proposition 3.12 [68] (by Theorem 2.1.12, Proposition 3.4(a) [68] and Remark 8.1.22).

8.1.4 Coherent quotients and topologies of finite type

In this section we shall study the lattice structure on the collection of coherent quotients of a given cartesian theory. Recall from section 2.2.4 that the notion of provability in a cartesian (resp. regular, coherent) theory $\mathbb{T}$ with respect to

cartesian (resp. regular, coherent) logic coincides with the notion of provability in $\mathbb{T}$ with respect to geometric logic.

Recall that a Grothendieck topology on $\mathcal{C}$ is said to be generated by a given collection $\mathcal{F}$ of presieves on $\mathcal{C}$ if it is the smallest Grothendieck topology J on $\mathcal{C}$ such that all the sieves generated by presieves in $\mathcal{F}$ are J -covering (cf. section 3.1.2).

Definition 8.1.23 A Grothendieck topology J on a category $\mathcal{C}$ is said to be of *finite type* if it is generated by a collection of finite presieves on $\mathcal{C}$.

Proposition 8.1.24 Let $\mathcal{C}$ be a category and J a Grothendieck topology on $\mathcal{C}$. Then J is of finite type if and only if there exists an assignment K sending to each object $c \in \mathcal{C}$ a collection $K(c)$ of finite presieves in $\mathcal{C}$ on c which satisfies the following properties:

- (i) If $R \in K(c)$ then for any arrow $g : d \rightarrow c$ in $\mathcal{C}$ there exists a presieve $S \in K(d)$ such that, for each arrow f in S , $g \circ f \in R$.
- (ii) If $\{f_i : c_i \rightarrow c \mid i \in I\} \in K(c)$ and for each $i \in I$ we have a presieve $\{g_{ij} : d_{ij} \rightarrow c_i \mid j \in J_i\} \in K(c_i)$ then there exists a presieve $P \in K(c)$ such that $P \subseteq \{f_i \circ g_{ij} : d_{ij} \rightarrow c \mid i \in I, j \in J_i\}$.
- (iii) For any sieve S on $c \in \mathcal{C}$, $S \in J(c)$ if and only if $S \supseteq T$ for some $T \in K(c)$.

Proof The ‘if’ part of the proposition follows immediately from Definition 3.1.3. Let us prove the ‘only if’ part. We define K by stipulating that, for any presieve V on an object $c \in \mathcal{C}$, $V \in K(c)$ if and only if V is finite and the sieve generated by it is J -covering. By Definition 3.1.3, K satisfies properties (i) and (ii) of the proposition. Let us now define K' as follows: for any sieve R on an object $c \in \mathcal{C}$, $R \in K'(c)$ if and only if $R \supseteq T$ for some $T \in K(c)$. We want to prove that $J = K'$. By Definition 3.1.3, K' is a Grothendieck topology and, clearly, K' is contained in J . But the fact that J is of finite type implies that $J \subseteq K'$, whence $J = K'$, as required. $\square$

Remark 8.1.25 The proof of Proposition 8.1.24 shows that a Grothendieck topology J on a small category $\mathcal{C}$ is of finite type if and only if every J -covering sieve contains a J -covering sieve generated by a finite presieve.

Proposition 8.1.26 Let $\mathcal{C}$ be a small category and J_1, J_2 Grothendieck topologies on $\mathcal{C}$.

- (i) If J_1, J_2 are of finite type then $J_1 \wedge J_2$ is of finite type.
- (ii) If J_1, J_2 are of finite type then $J_1 \vee J_2$ is of finite type.

Proof

- (i) For any sieve S on an object $c \in \mathcal{C}$, $S \in (J_1 \wedge J_2)(c)$ if and only if $S \in J_1(c)$ and $S \in J_2(c)$. Let us denote by K_1 (resp. K_2) the collection of finite presieves which generate a J_1 -covering (resp. a J_2 -covering) sieve, and let K be defined as follows: for any presieve V on $c \in \mathcal{C}$, $V \in K(c)$ if and only if there exist $V_1 \in K_1(c)$ and $V_2 \in K_2(c)$ such that $V = V_1 \cup V_2$. It

is immediate to see that K satisfies conditions (i) and (ii) of Proposition 8.1.24. Further, for any sieve S on $c \in \mathcal{C}$, $S \in (J_1 \wedge J_2)(c)$ if and only if $S \supseteq T$ for some $T \in K(c)$. So $J_1 \wedge J_2$ is of finite type by Proposition 8.1.24.

- (ii) Since $J_1 \vee J_2$ is the smallest Grothendieck topology on $\mathcal{C}$ which contains both J_1 and J_2 , our thesis follows at once by observing that we can get a collection of finite presieves generating $J_1 \vee J_2$ by taking the union of any two collections of finite presieves generating J_1 and J_2 . $\square$

Notice that, given two representations $\mathbf{Sh}(\mathcal{C}, J) \simeq \mathbf{Sh}(\mathcal{C}', J')$ of the same Grothendieck topos in terms of small sites, we may construct a bijection between the class $\mathfrak{Groth}_J^{\mathcal{C}}$ of Grothendieck topologies on $\mathcal{C}$ which contain J and the class $\mathfrak{Groth}_{J'}^{\mathcal{C}'}$ of Grothendieck topologies on $\mathcal{C}'$ which contain J' . Indeed, it is well known (cf. section 1.3.4) that the Grothendieck topologies on $\mathcal{C}$ (resp. $\mathcal{C}'$) which contain J (resp. J') are in bijection with the (isomorphism classes of) geometric inclusions into the topos $\mathbf{Sh}(\mathcal{C}, J)$ (resp. $\mathbf{Sh}(\mathcal{C}', J')$), whence composing geometric inclusions with the equivalence $\mathbf{Sh}(\mathcal{C}, J) \simeq \mathbf{Sh}(\mathcal{C}', J')$ yields a bijection between $\mathfrak{Groth}_J^{\mathcal{C}}$ and $\mathfrak{Groth}_{J'}^{\mathcal{C}'}$. Since the natural order between geometric inclusions corresponds to the canonical order between the associated Grothendieck topologies, our bijection between $\mathfrak{Groth}_J^{\mathcal{C}}$ and $\mathfrak{Groth}_{J'}^{\mathcal{C}'}$ is order-preserving and hence a Heyting algebra isomorphism. This fact can be notably exploited for obtaining explicit descriptions of the lattice operations between theories, as will be illustrated by the example in section 8.1.5. For instance, if $\mathbb{T}$ is a theory of presheaf type then its classifying topos can be represented both as $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ and as the presheaf topos $[\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}]$; thus, by Theorem 3.2.5, there is a Heyting algebra isomorphism

$$\mathfrak{Th}_{\Sigma}^{\mathbb{T}} \cong \mathfrak{Groth}_T^{\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}}$$

(where T is the trivial Grothendieck topology on $\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}$) between the (syntactic-equivalence classes of) quotients $\mathbb{T}'$ of $\mathbb{T}$ and the Grothendieck topologies J on the category $\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}$, with the property that for any such $\mathbb{T}'$ the topos $\mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}, J)$ of sheaves with respect to the corresponding Grothendieck topology J classifies $\mathbb{T}'$ (cf. section 8.1.2):

$$\begin{array}{ccc} & \text{Lattice of subtoposes of} & \\ & [\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) & \\ \text{Lattice structure on} & \text{---} & \text{Lattice structure on} \\ \mathfrak{Groth}_T^{\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}} & & \mathfrak{Th}_{\Sigma}^{\mathbb{T}} \end{array}$$

Proposition 8.1.27 *Let $\mathbb{T}$ be a cartesian theory over a signature Σ and $\mathcal{C}_{\mathbb{T}}^{\mathrm{cart}}$ its cartesian syntactic category. Then the bijection between the (syntactic-equivalence*

classes of) quotients of $\mathbb{T}$ and the Grothendieck topologies on $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$ induced by Theorem 3.2.5 restricts to a bijection between the (syntactic-equivalence classes of) coherent quotients of $\mathbb{T}$ and the finite-type Grothendieck topologies on $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$.

Proof The equivalence of classifying toposes $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \simeq [\mathcal{C}_{\mathbb{T}}^{\text{cart}}, \mathbf{Set}]$ induces by Theorem 3.2.5 a bijection between the (syntactic-equivalence classes of) geometric quotients of $\mathbb{T}$ and the Grothendieck topologies on $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$, which we may describe as follows. Given a Grothendieck topology J on $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$, the corresponding theory is axiomatized by the geometric sequents over Σ of the form $(\psi \vdash_{\vec{y}} \bigvee_{i \in I} (\exists \vec{x}_i) \theta_i)$, where $\{[\theta_i] \mid i \in I\}$ is any family of morphisms

$$\{\vec{x}_i . \phi_i\} \xrightarrow{[\theta_i]} \{\vec{y} . \psi\}$$

in $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$ forming a J -covering sieve. Conversely, by Proposition D1.3.10 [55], any geometric (resp. coherent) theory over Σ can be axiomatized by sequents of the form $(\psi \vdash_{\vec{y}} \bigvee_{i \in I} (\exists \vec{x}_i) \theta_i)$, where ψ and the θ_i are cartesian formulae over Σ such that for any $i \in I$ the sequent $(\theta_i \vdash_{\vec{x}_i, \vec{y}} \psi)$ is provable in geometric logic (where I may be taken finite if $\mathbb{T}$ is coherent); the corresponding Grothendieck topology on $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$ is thus generated by the sieves

$$\{\vec{x}_i, \vec{y}^{\vec{}} . \theta_i\} \xrightarrow{[\theta_i \wedge \vec{y} = \vec{y}^{\vec{}}]} \{\vec{y}^{\vec{}} . \psi\}$$

as i varies in I .

Our thesis then follows from Remark 3.2.9 or from Theorem 8.1.10 and Remark 8.1.11(b). $\square$

Remark 8.1.28 The analogue of Proposition 8.1.27 for regular theories also holds; specifically, the regular quotients of a cartesian theory $\mathbb{T}$ correspond to the Grothendieck topologies J on $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$ such that every J -covering sieve contains a J -covering sieve generated by a single arrow.

Corollary 8.1.29 *Let $\mathbb{T}$ be a cartesian theory over a signature Σ . Then the collection of (syntactic-equivalence classes of) coherent quotients of $\mathbb{T}$ form a sublattice of the collection $\mathfrak{Th}_{\Sigma}^{\mathbb{T}}$ of (syntactic-equivalence classes of) quotients of $\mathbb{T}$.*

Proof This follows immediately from Proposition 8.1.27, Theorem 3.2.5 and Proposition 8.1.26. $\square$

Corollary 8.1.29 implies that, more generally, for any geometric theory $\mathbb{T}$ over a signature Σ , the class of coherent theories in $\mathfrak{Th}_{\Sigma}^{\mathbb{T}}$ is closed under finite meets and joins in $\mathfrak{Th}_{\Sigma}^{\mathbb{T}}$; indeed, by the discussion preceding Proposition 8.1.27, the meets and joins of subtoposes of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{O}_{\Sigma}}, J_{\mathbb{T}}^{\mathbb{O}_{\Sigma}})$ (where $\mathbb{O}_{\Sigma}$ is the empty theory over Σ) are the same as those calculated in the lattice of subtoposes of $\mathbf{Sh}(\mathcal{C}_{\mathbb{O}_{\Sigma}}, J_{\mathbb{O}_{\Sigma}})$.

Remark 8.1.30 By Theorem 2.2.2, the order relation between coherent theories in $\mathfrak{Th}_{\Sigma}^{\mathbb{T}}$ is equivalent to the natural ordering on them defined as follows: $\mathbb{T}_1 \leq \mathbb{T}_2$ if and only if every (coherent) axiom of $\mathbb{T}_1$ is provable in $\mathbb{T}_2$ using coherent logic. Moreover, assuming the axiom of choice, the classical completeness theorem for coherent logic (Theorem 1.4.18) implies that this order relation also coincides with the well-known ordering on first-order theories, the condition $\mathbb{T}_1 \leq \mathbb{T}_2$ being equivalent to the requirement ‘for any Σ -structure M in **Set**, if M is a $\mathbb{T}_2$ -model then M is a $\mathbb{T}_1$ -model’.

8.1.5 An example

In this section, as an application of the results obtained in section 8.1.4, we calculate the meet of the theory of local rings and that of integral domains in the lattice of coherent (equivalently, of geometric) theories over the signature of the theory of commutative rings with unit.

Let Σ be the one-sorted signature consisting of two binary function symbols $+$ and $\cdot$, one unary function symbol $-$ and two constants 0 and 1 , and let $\mathbb{T}$ be the algebraic theory over Σ axiomatizing the notion of commutative ring with unit. The category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ coincides with the category $\mathbf{Rng}_{\text{f.g.}}$ of finitely generated commutative rings with unit. The theory $\mathbb{T}_1$ of local rings is obtained from $\mathbb{T}$ by adding the sequents

$$(0 = 1 \vdash_{\square} \perp)$$

and

$$((\exists z)((x + y) \cdot z = 1) \vdash_{x,y} (\exists z)(x \cdot z = 1) \vee (\exists z)(y \cdot z = 1)),$$

while the theory $\mathbb{T}_2$ of integral domains is obtained from $\mathbb{T}$ by adding the sequents

$$(0 = 1 \vdash_{\square} \perp)$$

and

$$(x \cdot y = 0 \vdash_{x,y} (x = 0) \vee (y = 0)).$$

Consider the Grothendieck topologies J_1 and J_2 on $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ corresponding respectively to $\mathbb{T}_1$ and $\mathbb{T}_2$. By Example D3.1.11(a) [55] and the calculation preceding Lemma 6.6 [13], we have the following descriptions of J_1 and J_2 : for any $A \in \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ and any sieve S on A in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$,

- $S \in J_1(A)$ if and only if S contains a finite family $\{\xi_i : A \rightarrow A[s_i^{-1}] \mid 1 \leq i \leq n\}$ of canonical inclusions $\xi_i : A \rightarrow A[s_i^{-1}]$ in $\mathbf{Rng}_{\text{f.g.}}$, where $\{s_1, \dots, s_n\}$ is any set of elements of A which is not contained in any proper ideal of A ;
- $S \in J_2(A)$ if and only if either A is the zero ring and S is the empty sieve on it or S contains a non-empty finite family $\{\pi_{a_i} : A \rightarrow A/(a_i) \mid 1 \leq i \leq n\}$ of canonical projections $\pi_{a_i} : A \rightarrow A/(a_i)$ in $\mathbf{Rng}_{\text{f.g.}}$, where $\{a_1, \dots, a_n\}$ is any set of elements of A such that $a_1 \cdot \dots \cdot a_n = 0$.

We may identify the polynomials with integer coefficients in a finite number of variables with the R -equivalence classes of terms over Σ , where R is the equivalence relation given by ' $t_1 R t_2$ if and only if $(\top \vdash t_1 = t_2)$ is provable in $\mathbb{T}$ '.

The meet $\mathbb{T}_1 \wedge \mathbb{T}_2$ in $\mathfrak{Th}_{\Sigma}^{\top}$ (or, equivalently, in $\mathfrak{Th}_{\Sigma}^{\text{Op}}$) corresponds to the intersection $J_1 \wedge J_2$ in $\mathbf{Groth}_T^{\mathbf{Rng}_{f.g.}^{\text{Op}}}$, where T is the trivial Grothendieck topology on $\mathbf{Rng}_{f.g.}^{\text{Op}}$ (cf. the discussion preceding Proposition 8.1.27). Now, the topology $J_1 \wedge J_2$ is generated by the collection of presieves of the form $V_1 \cup V_2$ where V_1 and V_2 are finite presieves on the same object of $\mathbf{Rng}_{f.g.}$ which respectively generate a J_1 -covering sieve and a J_2 -covering sieve (cf. the proof of Proposition 8.1.26(i)), and any object of $\mathbf{Rng}_{f.g.}$ has the form $\mathbb{Z}[x_1, \dots, x_n]/(P_1, \dots, P_s)$ where $P_1, \dots, P_s$ are polynomials in $\mathbb{Z}[x_1, \dots, x_n]$. It thus follows from Theorem 8.1.12 and Remark 8.1.11(b) that $\mathbb{T}_1 \wedge \mathbb{T}_2$ is the theory obtained from $\mathbb{T}$ by adding the sequents

$$(0 = 1 \vdash_{\square} \perp)$$

and

$$\left(\bigwedge_{1 \leq s \leq m} P_s(\vec{x}) = 0 \vdash_{\vec{x}} \bigvee_{1 \leq i \leq k} (\exists y)(G_i(\vec{x}) \cdot y = 1) \vee \bigvee_{1 \leq j \leq l} H_j(\vec{x}) = 0 \right),$$

for any polynomials P_s , G_i and H_j in a finite string $\vec{x}$ of variables with the property that if $\vec{x} = (x_1, \dots, x_n)$ then $\{P_1, \dots, P_s, G_1, \dots, G_k\}$ is any set of elements of $\mathbb{Z}[x_1, \dots, x_n]$ which is not contained in any proper ideal of $\mathbb{Z}[x_1, \dots, x_n]$ and $\prod_{1 \leq j \leq l} H_j \in (P_1, \dots, P_s)$ in $\mathbb{Z}[x_1, \dots, x_n]$.

8.2 Presheaf-type quotients

In this section we shall prove some results which show that it is often useful, in investigating whether a given geometric theory $\mathbb{S}$ is of presheaf type, to consider it in relation to a theory $\mathbb{T}$ of presheaf type of which $\mathbb{S}$ is a quotient.

8.2.1 Finality conditions

The following result is a corollary of Theorem 7.2.3 and Proposition 7.2.5.

Corollary 8.2.1 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ and $\mathbb{S}$ a quotient of $\mathbb{T}$ such that all the finitely presentable $\mathbb{S}$ -models are finitely presentable as $\mathbb{T}$ -models (e.g. $\mathbb{T}$ can be the empty theory over a finite signature and $\mathbb{S}$ any geometric theory over Σ whose finitely presentable models are all finite, cf. Proposition 9.1.1 below). Suppose, moreover, that for any object E of a Grothendieck topos $\mathcal{E}$ and any Σ -structure homomorphism $x : c \rightarrow \text{Hom}_{\mathcal{E}}(E, M)$, where c is a finitely presentable $\mathbb{T}$ -model and M is a $\mathbb{S}$ -model in $\mathcal{E}$, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$ -model homomorphism $f_i : c \rightarrow c_i$, where c_i is a finitely presentable $\mathbb{S}$ -model, and a Σ -structure homomorphism $x_i : c_i \rightarrow \text{Hom}_{\mathcal{E}}(E_i, M)$ such that $x_i \circ f_i = \text{Hom}_{\mathcal{E}}(e_i, M) \circ x$ for all $i \in I$. Then $\mathbb{S}$ is of presheaf type.*

Proof Condition (iii) of Theorem 6.3.1 is satisfied by Proposition 6.3.12, while the fact that conditions (i) and (ii) of Theorem 6.3.1 hold follows from Theorem 7.2.3 in view of Proposition 7.2.5 (in its statement take $\mathbb{T}$ equal to $\mathbb{S}$, $\mathbb{T}'$ equal to $\mathbb{T}$, $\mathcal{K}$ equal to the category of finitely presentable $\mathbb{S}$ -models and $\mathcal{H}$ equal to the category of finitely presentable $\mathbb{T}$ -models – the fact that $\int a$ is $\mathcal{E}$ -full readily follows from the fact that $\mathbb{S}$ is a quotient of $\mathbb{T}$). $\square$

In the sequel we shall refer to a family of homomorphisms x_i as in the statement of Corollary 8.2.1 as a ‘localization’ of the homomorphism x .

- Remarks 8.2.2** (a) Corollary 8.2.1, when applied to the empty theory $\mathbb{T}$ over a finite signature and to a geometric theory $\mathbb{S}$ over Σ whose finitely presentable models are all finite, generalizes Theorem 3.6 [73], whose hypotheses expressed syntactically in terms of geometric logic, once interpreted in topos-theoretic semantics, are stronger than those of the corollary.
- (b) Assuming the first of the hypotheses of Corollary 8.2.1, the second hypothesis is not only sufficient but also necessary for $\mathbb{S}$ to be of presheaf type. Indeed, if $\mathbb{S}$ is of presheaf type then every model M of $\mathbb{S}$ in a Grothendieck topos $\mathcal{E}$ is canonically an $\mathcal{E}$ -filtered colimit of a diagram of ‘constant’ models associated with the finitely presentable $\mathbb{S}$ -models (cf. sections 6.2.3 and 6.3.3), which by the first hypothesis are also finitely presentable as $\mathbb{T}$ -models, whence the fact that the second hypothesis of the corollary is satisfied follows from Propositions 6.2.13 and 6.2.15.
- (c) Corollary 8.2.1 can often be applied to pairs of the form $(\mathbb{S}, \mathbb{T} = \mathbb{S}_c)$, where $\mathbb{S}$ is a geometric theory over a signature Σ and $\mathbb{T}$ is equal to its cartesianization $\mathbb{S}_c$. It is not known if it is true in general that every finitely presentable model of a theory of presheaf type $\mathbb{T}$ is finitely presentable also as a $\mathbb{T}_c$ -model. Nonetheless, this property is satisfied by all the examples of theories of presheaf type considered in this book (see Chapter 9 below).

Corollary 8.2.1 can also be applied to pairs consisting of a geometric theory over a signature Σ and the empty theory over the same signature, provided that the former satisfies appropriate hypotheses. In order to apply the corollary in this context, we need the following lemma. Below in this section, given a finite signature Σ , when we say that a Σ -structure c is *finite* we mean that cA is finite for every sort A of Σ .

Lemma 8.2.3 *Let Σ be a finite signature and $\mathbb{T}$ a geometric theory over Σ with a finite number of axioms each of which is of the form $(\top \vdash_{\bar{x}} \bigvee_{i \in I} \phi_i)$, where the ϕ_i are finite conjunctions of atomic formulae. Then, for any $\mathbb{T}$ -model M in a Grothendieck topos $\mathcal{E}$, any object E of $\mathcal{E}$ and any Σ -structure homomorphism $f : c \rightarrow \text{Hom}_{\mathcal{E}}(E, M)$, where c is a finite Σ -structure, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a finite Σ -substructure c_i of $\text{Hom}_{\mathcal{E}}(E_i, M)$ which is a model of $\mathbb{T}$ and a Σ -structure homomorphism $f_i : c \rightarrow c_i$ such that $\text{Hom}_{\mathcal{E}}(e_i, M) \circ f = j_i \circ f_i$ (where j_i is the canonical inclusion of c_i into $\text{Hom}_{\mathcal{E}}(E_i, M)$).*

Proof For each axiom σ of the form $(\top \vdash_{x_\sigma} \bigvee_{i \in I_\sigma} \phi_i^\sigma)$ and any finite tuple $\vec{\xi}$ of elements of c of the same type as x_σ , since M is by our hypotheses a $\mathbb{T}$ -model, there exists an epimorphic family $\{e_i^\xi : E_i^\xi \rightarrow E \mid i \in I_\sigma\}$ in $\mathcal{E}$ such that for each $i \in I_\sigma$, $f(\vec{\xi}) \circ e_i^\xi$ factors through $[[x_\sigma \cdot \phi_i]]_M$. Since there is only a finite number of axioms of $\mathbb{T}$ and of elements of c , and the fibred product of a finite number of epimorphic families is again an epimorphic family, there exists an epimorphic family $\{e_k : E_k \rightarrow E \mid k \in K\}$ in $\mathcal{E}$ such that, for any $k \in K$, any axiom σ of $\mathbb{T}$ and any tuple $\vec{\xi}$ of elements of c of the appropriate sorts, the element $f(\vec{\xi}) \circ e_k \in \text{Hom}_{\mathcal{E}}(E_k, M)$ satisfies the formula on the right-hand side of the sequent σ . Hence if we consider, for each $k \in K$, the surjection-inclusion factorization $j_k \circ f_k$ of the homomorphism $\text{Hom}_{\mathcal{E}}(e_k, M) \circ f : c \rightarrow \text{Hom}_{\mathcal{E}}(E_k, M)$ (in the sense of Lemma 7.2.37), we obtain that $c_k = \text{Im}(f_k)$ is finite, since f_k is surjective, and that all the axioms of $\mathbb{T}$ are satisfied in c_k (since the formulae ϕ_i are finite conjunctions of atomic formulae and c_k is a Σ -substructure of $\text{Hom}_{\mathcal{E}}(E_k, M)$). This proves our thesis. $\square$

Corollary 8.2.4 *Let Σ be a finite signature and $\mathbb{T}$ a geometric theory over Σ with a finite number of axioms each of which is of the form $(\top \vdash_{\vec{x}} \bigvee_{i \in I} \phi_i)$, where the ϕ_i are finite conjunctions of atomic formulae. Suppose that for any $\mathbb{T}$ -model M in a Grothendieck topos $\mathcal{E}$ and any object E of $\mathcal{E}$, every finitely generated Σ -substructure of $\text{Hom}_{\mathcal{E}}(E, M)$ has sortwise only a finite number of elements besides the constants (e.g. when the signature Σ does not contain function symbols except for a finite number of constants). Then $\mathbb{T}$ is of presheaf type, classified by the topos of covariant set-valued functors on the category of finite models of $\mathbb{T}$.*

Proof As Σ is finite, every $\mathbb{T}$ -model that sortwise contains only finitely many elements besides the constants is finitely presentable as a model of the empty theory $\mathbb{O}_\Sigma$ over Σ (cf. Proposition 9.1.1 below). This remark ensures, by Lemma 8.2.3, that the theory $\mathbb{T}$ satisfies the hypotheses of Corollary 8.2.1 with respect to $\mathbb{O}_\Sigma$. Since the latter theory is of presheaf type (as it is cartesian), it follows that $\mathbb{T}$ is of presheaf type as well, as required. $\square$

8.2.2 Rigid topologies

In this section we shall analyse the presheaf-type quotients of a theory of presheaf type $\mathbb{T}$ in terms of the associated subtoposes of the classifying topos of $\mathbb{T}$.

Recall from section 6.1.3 that a Grothendieck topology J on a category $\mathcal{C}$ is said to be rigid if for every object c of $\mathcal{C}$, the family of all the arrows to c in $\mathcal{C}$ from J -irreducible objects of $\mathcal{C}$ (i.e. objects d with the property that the only J -covering sieve on d is the maximal one), generates a J -covering sieve.

Let us consider a theory of presheaf type $\mathbb{T}$ and a quotient $\mathbb{T}'$ of $\mathbb{T}$ corresponding to a Grothendieck topology J on $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ as in section 8.1.2. The following result gives a characterization of the models of $\mathbb{T}'$ which are finitely

presentable as models of $\mathbb{T}$ in terms of the topology J ; it is useful for recognizing the quotients of a theory of presheaf type which are again of presheaf type.

Theorem 8.2.5 *Let $\mathbb{T}'$ be a quotient of a theory of presheaf type $\mathbb{T}$ corresponding via the duality of Theorem 3.2.5 to a Grothendieck topology J on $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ (as in section 8.1.2), and let M be a finitely presentable $\mathbb{T}$ -model. Then M is a model of $\mathbb{T}'$ if and only if it is J -irreducible as an object of $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$.*

Proof As shown in section 8.1.2, the theory $\mathbb{T}'$ can be identified with the quotient of $\mathbb{T}$ obtained by adding all the sequents σ of the form $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} (\exists \vec{y}_i) \theta_i)$ where, for any $i \in I$, $[\theta_i] : \{\vec{y}_i \cdot \psi_i\} \rightarrow \{\vec{x} \cdot \phi\}$ is an arrow in $\mathcal{C}_{\mathbb{T}}$, $\phi(\vec{x})$ and $\psi_i(\vec{y}_i)$ are formulae presenting respectively $\mathbb{T}$ -models $M_{\{\vec{x} \cdot \phi\}}$ and $M_{\{\vec{y}_i \cdot \psi_i\}}$ and the sieve S_{σ} on $M_{\{\vec{x} \cdot \phi\}}$ in the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ generated by the arrows $M_{[\theta_i]}$ (using the notation in the proof of Theorem 6.1.13) is J -covering. On the other hand, the Grothendieck topology J associated with the quotient $\mathbb{T}'$ consists of all the sieves of the form S_{σ} for a sequent σ provable in $\mathbb{T}'$ of the form $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} (\exists \vec{y}_i) \theta_i)$, where, for any $i \in I$, $[\theta_i] : \{\vec{y}_i \cdot \psi_i\} \rightarrow \{\vec{x} \cdot \phi\}$ is an arrow in $\mathcal{C}_{\mathbb{T}}$ and $\phi(\vec{x})$, $\psi_i(\vec{y}_i)$ are formulae presenting respectively $\mathbb{T}$ -models $M_{\{\vec{x} \cdot \phi\}}$ and $M_{\{\vec{y}_i \cdot \psi_i\}}$.

Let us establish the following claim, from which our thesis will follow easily: for any sequent $\sigma := (\phi \vdash_{\vec{x}} \bigvee_{i \in I} (\exists \vec{y}_i) \theta_i)$ of the above form and any set-based $\mathbb{T}$ -model N , σ holds in N if and only if every arrow $f : M_{\{\vec{x} \cdot \phi\}} \rightarrow N$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$ factors through an arrow in S_{σ} .

Clearly, σ holds in N if and only if for any $\vec{a} \in [[\vec{x} \cdot \phi]]_N$ there exist $i \in I$ and $\vec{b} \in [[\vec{y}_i \cdot \psi_i]]_N$ such that $(\vec{b}, \vec{a}) \in [[\vec{y}_i, \vec{x} \cdot \theta_i]]_N$. By the universal property of the model $M_{\{\vec{x} \cdot \phi\}}$, the tuples $\vec{a} \in [[\vec{x} \cdot \phi]]_N$ can be identified with the arrows $M_{\{\vec{x} \cdot \phi\}} \rightarrow N$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$ via the correspondence which assigns to $\vec{a} \in [[\vec{x} \cdot \phi]]_N$ the unique $\mathbb{T}$ -model homomorphism $f_{\vec{a}} : M_{\{\vec{x} \cdot \phi\}} \rightarrow N$ such that $f_{\vec{a}}(\vec{\xi}_{\phi}) = \vec{a}$, and similarly for the tuples $\vec{b} \in [[\vec{y}_i \cdot \psi_i]]_N$. We can easily see that the property that $(\vec{b}, \vec{a}) \in [[\vec{y}_i, \vec{x} \cdot \theta_i]]_N$ is exactly equivalent to the requirement that $f_{\vec{b}} \circ M_{[\theta_i]} = f_{\vec{a}}$; indeed, the equality $f_{\vec{b}} \circ M_{[\theta_i]} = f_{\vec{a}}$ holds if and only if $(f_{\vec{b}} \circ M_{[\theta_i]})(\vec{\xi}_{\phi}) = f_{\vec{a}}(\vec{\xi}_{\phi})$ and, since $M_{[\theta_i]}(\vec{\xi}_{\phi}) = [[\vec{y}_i, \vec{x} \cdot \theta_i]]_{M_{\{\vec{y}_i \cdot \psi_i\}}}(\vec{\xi}_{\psi_i})$, we have that $(f_{\vec{b}} \circ M_{[\theta_i]})(\vec{\xi}_{\phi}) = f_{\vec{b}}([[\vec{y}_i, \vec{x} \cdot \theta_i]]_{M_{\{\vec{y}_i \cdot \psi_i\}}}(\vec{\xi}_{\psi_i})) = [[\vec{y}_i, \vec{x} \cdot \theta_i]]_N(f_{\vec{b}}(\vec{\xi}_{\psi_i})) = [[\vec{y}_i, \vec{x} \cdot \theta_i]]_N(\vec{b})$ (where we have identified $[[\vec{y}_i, \vec{x} \cdot \theta_i]]_{M_{\{\vec{y}_i \cdot \psi_i\}}}$ and $[[\vec{y}_i, \vec{x} \cdot \theta_i]]_N$ with the maps of which they are the graphs). This proves our claim.

Now, for any sequent σ of the above form which is provable in $\mathbb{T}'$, the associated sieve S_{σ} is J -covering whence for any arrow $f : M_{\{\vec{x} \cdot \phi\}} \rightarrow M$ in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ the pullback sieve $f^*(S_{\sigma})$ is J -covering on M and hence maximal (i.e. f belongs to S_{σ}) if M is J -irreducible. The above claim thus shows that if M is J -irreducible then M is a model of $\mathbb{T}'$. In the converse direction, if M is a model of $\mathbb{T}'$, which we can suppose of the form $M_{\{\vec{x} \cdot \phi\}}$ for some geometric formula $\phi(\vec{x})$ over the signature of $\mathbb{T}$ by Theorem 6.1.13, then for any J -covering

sieve $S := S_\sigma$ on $M = M_{\{\bar{x}, \phi\}}$ in the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ the sequent $\sigma := (\phi \vdash_{\bar{x}} \bigvee_{i \in I} (\exists \bar{y}_i) \theta_i)$ is provable in $\mathbb{T}'$ and hence valid in M , so by the above claim the identity on M belongs to S_σ , i.e. S_σ is maximal. $\square$

We can characterize the Grothendieck topologies J such that the corresponding subtopos $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J) \hookrightarrow [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ is an essential geometric inclusion (that is, a geometric inclusion whose inverse image admits a left adjoint) by using the following site characterizations:

- (1) A canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$, where $\mathcal{C}$ is a small Cauchy-complete category and the topos $\mathbf{Sh}(\mathcal{C}, J)$ is equivalent to a presheaf topos, is essential if and only if the Grothendieck topology J is rigid.
- (2) A geometric morphism (resp. a geometric inclusion) $[\mathcal{C}, \mathbf{Set}] \rightarrow [\mathcal{D}, \mathbf{Set}]$, where $\mathcal{D}$ is a Cauchy-complete category, is essential if and only if it is induced by a functor (resp. by a full and faithful functor) $\mathcal{C} \rightarrow \mathcal{D}$.

The latter characterization is well-known (cf. Lemma A4.1.5 and Example A4.2.12(b) [55]), while the former can be proved as follows. If J is rigid then, denoting by $\mathcal{D}$ the full subcategory of $\mathcal{C}$ on the J -irreducible objects, the Comparison Lemma yields an equivalence $\mathbf{Sh}(\mathcal{C}, J) \simeq [\mathcal{D}^{\text{op}}, \mathbf{Set}]$ which makes the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ isomorphic to the canonical geometric inclusion $[\mathcal{D}^{\text{op}}, \mathbf{Set}] \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ induced by the full embedding of categories $\mathcal{D}^{\text{op}} \hookrightarrow \mathcal{C}^{\text{op}}$; in particular, the geometric morphism $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is essential. Conversely, if the canonical geometric inclusion $i_J : \mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is essential and $\mathbf{Sh}(\mathcal{C}, J)$ is equivalent to a presheaf topos $[\mathcal{D}^{\text{op}}, \mathbf{Set}]$, by (2) we have that i_J is isomorphic to a geometric inclusion $[\mathcal{D}^{\text{op}}, \mathbf{Set}] \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ induced by a full embedding $\mathcal{D}^{\text{op}} \hookrightarrow \mathcal{C}^{\text{op}}$. Now, the topology $J_{\mathcal{D}}$ on $\mathcal{C}$ defined by saying that a sieve R on an object c of $\mathcal{C}$ is J -covering if and only if it contains all the morphisms from objects of $\mathcal{D}$ to c is clearly rigid, and the Comparison Lemma yields an equivalence $\mathbf{Sh}(\mathcal{C}, J_{\mathcal{D}}) \simeq [\mathcal{D}^{\text{op}}, \mathbf{Set}]$ which makes the geometric morphism $[\mathcal{D}^{\text{op}}, \mathbf{Set}] \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ isomorphic to the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, J_{\mathcal{D}}) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ (cf. Example A2.2.4(d) [55]). It then follows that $J = J_{\mathcal{D}}$; in particular, J is rigid.

These site characterizations define the arches of a ‘bridge’ (in the sense of section 2.2) leading to the following result:

Theorem 8.2.6 *Let $\mathbb{T}'$ be a quotient of a theory of presheaf type $\mathbb{T}$, corresponding to a Grothendieck topology J on the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ under the duality of Theorem 3.2.5 (as in section 8.1.2). Then the topology J is rigid if and only if $\mathbb{T}'$ is of presheaf type and every finitely presentable $\mathbb{T}'$ -model is finitely presentable also as a $\mathbb{T}$ -model.*

Proof If $\mathbb{T}'$ is of presheaf type and every finitely presentable $\mathbb{T}'$ -model is finitely presentable also as a $\mathbb{T}$ -model then the geometric inclusion corresponding via the duality of Theorem 3.2.5 to the quotient $\mathbb{T}'$ of $\mathbb{T}$ is induced by the embedding

$$\text{f.p.}\mathbb{T}'\text{-mod}(\mathbf{Set}) \hookrightarrow \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$$

(cf. Theorem 8.2.5). From the above characterizations, it thus follows that the Grothendieck topology J is rigid. Conversely, if J is rigid then $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J)$ is equivalent to the presheaf topos $[\mathcal{D}, \mathbf{Set}]$, where $\mathcal{D}$ is the opposite of the full subcategory of $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ on the J -irreducible objects. So $\mathbb{T}'$ is of presheaf type. Clearly, $\mathcal{D}$ is Cauchy-complete (as $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ is Cauchy-complete and the property of J -irreducibility is stable under retracts) and hence, since $\mathbb{T}'$ is of presheaf type, it is equal to the category of finitely presentable $\mathbb{T}'$ -models; in particular, every finitely presentable $\mathbb{T}'$ -model is finitely presentable as a $\mathbb{T}$ -model. $\square$

Remarks 8.2.7 (a) The proof of Theorem 8.2.6 shows that if $\mathbb{T}'$ is of presheaf type and every finitely presentable $\mathbb{T}'$ -model is finitely presentable as a $\mathbb{T}$ -model, the full subcategory $\text{f.p.}\mathbb{T}'\text{-mod}(\mathbf{Set})$ of $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ and the topology J can be defined in terms of each other as follows:

- the objects of $\text{f.p.}\mathbb{T}'\text{-mod}(\mathbf{Set})$ are precisely the J -irreducible objects of $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$;
 - a sieve S in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ on an object c is J -covering if and only if it contains all the arrows in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ from finitely presentable $\mathbb{T}'$ -models to c .
- (b) If $\mathbb{T}'$ is a presheaf-type quotient of a theory of presheaf type $\mathbb{T}$ then, for any finitely presentable $\mathbb{T}$ -model c , any $\mathbb{T}'$ -model M and any $\mathbb{T}$ -model homomorphism $f : c \rightarrow M$, there exist a $\mathbb{T}$ -model homomorphism $g : c \rightarrow c'$ to a finitely presentable $\mathbb{T}'$ -model c' and a $\mathbb{T}$ -model homomorphism $h : c' \rightarrow M$ such that $h \circ g = f$. Indeed, M can be expressed as a filtered colimit of finitely presentable $\mathbb{T}'$ -models c' ; therefore, as c is finitely presentable as a $\mathbb{T}$ -model, f necessarily factors through a colimit arrow $c' \rightarrow M$ (cf. Remark 8.2.2(b)).
- (c) Under the hypothesis that every finitely presentable $\mathbb{T}'$ -model is finitely presentable also as a $\mathbb{T}$ -model, the theory $\mathbb{T}'$ can be characterized as the quotient of $\mathbb{T}$ obtained by adding all the sequents of the form $(\phi(\vec{x}) \vdash_{\vec{x}} \bigvee_{i \in I} \exists \vec{x}_i \theta_i(\vec{x}_i, \vec{x}))$ where $\phi(\vec{x})$ is a $\mathbb{T}$ -irreducible formula (equivalently, a formula presenting a $\mathbb{T}$ -model) and $\{\theta_i(\vec{x}_i, \vec{x}) : \{\vec{x}_i \cdot \phi_i\} \rightarrow \{\vec{x} \cdot \phi\} \mid i \in I\}$ is the (set of representatives of formulae in the) family of $\mathbb{T}$ -provably functional formulae from a formula $\phi_i(\vec{x}_i)$ presenting a $\mathbb{T}'$ -model to $\phi(\vec{x})$ (cf. the proof of Theorem 8.2.5 and Remark 8.2.7(a)).
- (d) Theorem 8.2.6 has been applied in [32] (cf. Theorem 6.20 therein) to prove that the theory of local MV-algebras in a proper variety of MV-algebras is of presheaf type.

The following result identifies a natural class of presheaf-type quotients of presheaf-type theories satisfying the condition in Theorem 8.2.6.

Theorem 8.2.8 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ . Then any quotient $\mathbb{T}'$ of $\mathbb{T}$ obtained from $\mathbb{T}$ by adding sequents of the form $(\phi \vdash_{\vec{x}} \perp)$, where $\phi(\vec{x})$ is a geometric formula over Σ , is classified by the topos $[\mathcal{T}, \mathbf{Set}]$, where $\mathcal{T}$ is the full subcategory of $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ on the $\mathbb{T}'$ -models.*

Proof Since by Corollary 6.1.10 we can cover each $\phi(\vec{x})$ with $\mathbb{T}$ -irreducible formulae in the syntactic category $\mathcal{C}_{\mathbb{T}}$ of $\mathbb{T}$, we can suppose without loss of generality all the $\phi(\vec{x})$ to be $\mathbb{T}$ -irreducible, that is to present a $\mathbb{T}$ -model $M_{\{\vec{x}, \phi\}}$. Then, by Theorem 8.1.10, $\mathbb{T}'$ is classified by the topos $\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J)$, where J is the smallest Grothendieck topology on $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ which contains all the empty sieves on the models $M_{\{\vec{x}, \phi\}}$ presented by the formulae $\phi(\vec{x})$ occurring in the axioms $(\phi \vdash_{\vec{x}} \perp)$ added to $\mathbb{T}$ to form $\mathbb{T}'$. Let $\mathcal{T}$ be the full subcategory of $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ on the $\mathbb{T}'$ -models. Then $\mathcal{T}$ is J -dense; indeed, any object not in $\mathcal{T}$ admits an arrow in $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ from a model of the form $M_{\{\vec{x}, \phi\}}$ and hence is J -covered by the empty sieve in $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$. Further, for any object of $\mathcal{T}$, the J -covering sieves on it are exactly the maximal ones; therefore, the Comparison Lemma yields an equivalence $\mathbf{Sh}(\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J) \simeq [\mathcal{T}, \mathbf{Set}]$, as required. $\square$

Remark 8.2.9 Theorem 8.2.8 could alternatively be deduced from Corollary 8.2.1 by observing that if $\mathbb{T}$ and $\mathbb{T}'$ are theories as in the hypotheses of the theorem then the domain of any $\mathbb{T}$ -model homomorphism having as codomain a $\mathbb{T}'$ -model is also a $\mathbb{T}'$ -model whence the hypotheses of the corollary are trivially satisfied.

8.2.3 Finding theories classified by a given presheaf topos

The following theorem provides a method for constructing theories of presheaf type whose category of finitely presentable models is equivalent, up to Cauchy completion, to a given small category of structures.

Theorem 8.2.10 *Let $\mathbb{T}$ be a theory of presheaf type and $\mathcal{A}$ a full subcategory of $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$. Then the $\mathcal{A}$ -completion $\mathbb{T}_{\mathcal{A}}$ of $\mathbb{T}$ (i.e. the theory consisting of all the geometric sequents over the signature of $\mathbb{T}$ which are valid in all the models in $\mathcal{A}$) is of presheaf type classified by the topos $[\mathcal{A}, \mathbf{Set}]$; in particular, the presentable $\mathbb{T}_{\mathcal{A}}$ -models are precisely the retracts of models in $\mathcal{A}$.*

Proof Since every model in $\mathcal{A}$ is finitely presentable as a $\mathbb{T}$ -model, we have a geometric inclusion $i : [\mathcal{A}, \mathbf{Set}] \hookrightarrow \mathbf{Set}[\mathbb{T}] \simeq [\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ induced by the canonical inclusion $\mathcal{A} \hookrightarrow \mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$. This subtopos corresponds, by the duality of Theorem 3.2.5, to a quotient of $\mathbb{T}$ classified by the topos $[\mathcal{A}, \mathbf{Set}]$, which can be characterized as the collection of geometric sequents which hold in every model in $\mathcal{A}$, that is as the $\mathcal{A}$ -completion $\mathbb{T}_{\mathcal{A}}$ of $\mathbb{T}$ (cf. Remark 6.1.2(c)).

$$\begin{array}{ccccc}
& & [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \simeq & \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) & \\
& \nearrow & \uparrow & \uparrow & \searrow \\
& & [\mathcal{A}, \mathbf{Set}] & \simeq & \mathbf{Sh}(\mathcal{C}_{\mathbb{T}_{\mathcal{A}}}, J_{\mathbb{T}_{\mathcal{A}}}) \\
& \nwarrow & & & \searrow \\
\mathcal{A} & \downarrow & & & \downarrow \\
\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}) & & & & \mathbb{T}_{\mathcal{A}}
\end{array}$$

Therefore $\mathbb{T}_{\mathcal{A}}$ is of presheaf type classified by the topos $[\mathcal{A}, \mathbf{Set}]$. In particular, the finitely presentable $\mathbb{T}_{\mathcal{A}}$ -models are the finitely presentable objects of $\text{Ind-}\mathcal{A}$, that is the retracts of objects of $\mathcal{A}$ in $\text{Ind-}\mathcal{A} \simeq \mathbb{T}_{\mathcal{A}}\text{-mod}(\mathbf{Set})$. $\square$

- Remarks 8.2.11** (a) Theorem 8.2.10 is a generalization of Joyal-Wraith’s recognition theorem (Theorem 1.1 [5]), as well as of Proposition 5.3 [51]. Indeed, Theorem 1.1 [5] can be obtained as the particular case when $\mathbb{T}_{\mathcal{A}}$ has enough set-based models and every set-based model of $\mathbb{T}_{\mathcal{A}}$ is a filtered colimit of finitely presentable models of $\mathbb{T}_{\mathcal{A}}$ which are also finitely presentable as $\mathbb{T}$ -models, while Proposition 5.3 [51] can be obtained as the specialization of this result to the case when $\mathbb{T}_{\mathcal{A}}$ is a disjunctive theory and $\mathbb{T}$ is a cartesian theory (in view of the fact that, by Theorem 4.1 [51], the category of set-based models of a disjunctive theory $\mathbb{S}$ is multiply finitely presentable and hence every set-based model of $\mathbb{S}$ can be expressed as a filtered colimit of finitely presentable $\mathbb{S}$ -models).
- (b) It is worth comparing Theorem 8.2.10 with Theorem 6.3.1: if $\mathbb{T}'$ is a quotient of a theory of presheaf type $\mathbb{T}$ and $\mathcal{K}$ is a small category of set-based $\mathbb{T}'$ -models such that every model in $\mathcal{K}$ is finitely presentable as a $\mathbb{T}$ -model then, in order to conclude that $\mathbb{T}'$ is of presheaf type classified by the topos $[\mathcal{K}, \mathbf{Set}]$, we can either verify conditions (i) and (ii) of Theorem 6.3.1 or prove that the models in $\mathcal{K}$ are jointly conservative for $\mathbb{T}'$. Indeed, as observed in Remark 6.3.2(e), conditions (i) and (ii) of Theorem 6.3.1 together imply that the models in $\mathcal{K}$ are jointly conservative for $\mathbb{T}'$.

Corollary 8.2.12 *Assuming the axiom of choice, every coherent theory (or, more generally, any theory in a countable fragment of geometric logic as in the hypotheses of Theorem 5.1.7 [64]) over a finite relational signature whose axioms do not involve existential quantifications is of presheaf type.*

Proof Let $\mathbb{T}$ be a theory satisfying the hypotheses of the Corollary. Since the axioms of $\mathbb{T}$ do not involve existential quantifications, every substructure of a model of $\mathbb{T}$ is a model of $\mathbb{T}$. Moreover, since the signature of $\mathbb{T}$ is relational, every finitely generated structure over the signature of $\mathbb{T}$ contains only a finite number of elements besides the constants. So every finitely presentable model of $\mathbb{T}$ contains only a finite number of elements besides the constants (as any model of $\mathbb{T}$ is a filtered union of its finite submodels). On the other hand, since the signature of $\mathbb{T}$ is finite, every such model is finitely presentable as a model of the empty theory over the signature of $\mathbb{T}$ (cf. Proposition 9.1.1 below). Therefore condition

(iii) of Theorem 6.3.1 is satisfied by the theory $\mathbb{T}$ with respect to the category of its finite models and homomorphisms between them (cf. Proposition 6.3.12(i)). Now, assuming the axiom of choice, every theory satisfying the hypotheses of Theorem 5.1.7 [64], in particular any coherent theory, has enough set-based models, whence our thesis follows from Theorem 8.2.10. $\square$

In light of Theorem 8.2.10, a natural first step in constructing a geometric theory classified by a presheaf topos $[\mathcal{K}, \mathbf{Set}]$ consists in finding a theory of presheaf type $\mathbb{T}$ such that the category $\mathcal{K}$ can be identified with a full subcategory of the category of finitely presentable $\mathbb{T}$ -models. Indeed, under these hypotheses, Theorem 8.2.10 ensures the existence of a quotient $\mathbb{T}_{\mathcal{K}}$ of $\mathbb{T}$ classified by the topos $[\mathcal{K}, \mathbf{Set}]$, which can be characterized as the theory consisting of all the geometric sequents over the signature of $\mathbb{T}$ which are valid in every model in $\mathcal{K}$. In most cases, if one has a natural candidate $\mathbb{S}$ for a theory classified by the topos $[\mathcal{K}, \mathbf{Set}]$, the theory $\mathbb{T}$ can be chosen to be the Horn part of $\mathbb{S}$ (i.e. the theory consisting of all the Horn sequents which are provable in $\mathbb{S}$) or the cartesianization $\mathbb{S}_c$ of $\mathbb{S}$.

Of course, the abstract characterization of $\mathbb{T}_{\mathcal{K}}$ as the theory consisting of all the geometric sequents over the signature of $\mathbb{T}$ which are valid in every model in $\mathcal{K}$ is not very useful in specific contexts where one looks for an axiomatization of $\mathbb{T}_{\mathcal{K}}$ that is as simple and ‘economical’ as possible. To address this issue, we observe that if every $\mathbb{T}$ -model M in $\mathcal{K}$ is strongly finitely presented as a model of $\mathbb{T}$ as well as finitely generated (in the sense that for any sort A of the signature of $\mathbb{T}$ the elements of the set MA are precisely given by the interpretations in M of terms $t(\vec{x})$ whose output sort is A , where $\vec{x}$ are the generators of M as a strongly finitely presented model of $\mathbb{T}$) then we dispose of an explicit axiomatization of the theory $\mathbb{T}_{\mathcal{K}}$, as shown by the following result:

Theorem 8.2.13 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathcal{K}$ a full subcategory of the category of set-based $\mathbb{T}$ -models such that every $\mathbb{T}$ -model in $\mathcal{K}$ is both strongly finitely presentable and finitely generated (with respect to the same generators). Then the following sequents, where we denote by $\mathcal{P}$ the set of (representatives of) geometric formulae over Σ which strongly present a $\mathbb{T}$ -model in $\mathcal{K}$, added to the axioms of $\mathbb{T}$, yield an axiomatization of a quotient of $\mathbb{T}$ classified by the topos $[\mathcal{K}, \mathbf{Set}]$ via the Morita equivalence induced by the canonical geometric morphism $p_{\mathcal{K}} : [\mathcal{K}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$:*

(i) *the sequent*

$$\left(\top \vdash_{\square} \bigvee_{\phi(\vec{x}) \in \mathcal{P}} (\exists \vec{x}) \phi(\vec{x}) \right);$$

(ii) *for any formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$, where $\vec{x} = (x_1^{A_1}, \dots, x_n^{A_n})$ and $\vec{y} = (y_1^{B_1}, \dots, y_m^{B_m})$, the sequent*

$$\left(\phi(\vec{x}) \wedge \psi(\vec{y}) \vdash_{\vec{x}, \vec{y}} \bigvee_{\substack{\chi(\vec{z}) \in \mathcal{P}, \\ s_1^{B_1}(\vec{z}), \dots, s_m^{B_m}(\vec{z})}} (\exists \vec{z}) \left(\chi(\vec{z}) \wedge \bigwedge_{\substack{i \in \{1, \dots, n\}, \\ j \in \{1, \dots, m\}}} (x_i = t_i(\vec{z}) \wedge y_j = s_j(\vec{z})) \right) \right),$$

where the disjunction is taken over all the formulae $\chi(\vec{z})$ in $\mathcal{P}$ and all the tuples of terms $t_1^{A_1}(\vec{z}), \dots, t_n^{A_n}(\vec{z})$ and $s_1^{B_1}(\vec{z}), \dots, s_m^{B_m}(\vec{z})$ such that

$$(t_1(\vec{\xi}_\chi), \dots, t_n(\vec{\xi}_\chi)) \in [[\vec{x} \cdot \phi]]_{M_{\{\vec{z} \cdot \chi\}}}$$

and

$$(s_1(\vec{\xi}_\chi), \dots, s_m(\vec{\xi}_\chi)) \in [[\vec{y} \cdot \psi]]_{M_{\{\vec{z} \cdot \chi\}}};$$

- (iii) for any formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$, where $\vec{x} = (x_1^{A_1}, \dots, x_n^{A_n})$ and $\vec{y} = (y_1^{B_1}, \dots, y_m^{B_m})$, and any terms $t_1^{A_1}(\vec{y}), s_1^{A_1}(\vec{y}), \dots, t_n^{A_n}(\vec{y}), s_n^{A_n}(\vec{y})$ such that $(t_1(\vec{\xi}_\phi), \dots, t_n(\vec{\xi}_\phi)) \in [[\vec{y} \cdot \phi]]_{M_{\{\vec{y} \cdot \psi\}}}$ and $(s_1(\vec{\xi}_\phi), \dots, s_n(\vec{\xi}_\phi)) \in [[\vec{y} \cdot \phi]]_{M_{\{\vec{y} \cdot \psi\}}}$, the sequent

$$\left(\bigwedge_{i \in \{1, \dots, n\}} (t_i(\vec{y}) = s_i(\vec{y})) \wedge \psi(\vec{y}) \vdash_{\vec{y}} \bigvee_{\substack{\chi(\vec{z}) \in \mathcal{P}, \\ u_1^{B_1}(\vec{z}), \dots, u_m^{B_m}(\vec{z})}} (\exists \vec{z}) \left(\chi(\vec{z}) \wedge \bigwedge_{j \in \{1, \dots, m\}} (y_j = u_j(\vec{z})) \right) \right),$$

where the disjunction is taken over all the formulae $\chi(\vec{z})$ in $\mathcal{P}$ and all the tuples of terms $u_1^{B_1}(\vec{z}), \dots, u_m^{B_m}(\vec{z})$ such that

$$(u_1(\vec{\xi}_\chi), \dots, u_m(\vec{\xi}_\chi)) \in [[\vec{y} \cdot \psi]]_{M_{\{\vec{z} \cdot \chi\}}}$$

and

$$t_i(u_1(\vec{\xi}_\chi), \dots, u_m(\vec{\xi}_\chi)) = s_i(u_1(\vec{\xi}_\chi), \dots, u_m(\vec{\xi}_\chi)) \text{ in } M_{\{\vec{z} \cdot \chi\}}$$

for all $i \in \{1, \dots, n\}$;

- (iv) for any sort A of Σ , the sequent

$$\left(\top \vdash_{x_A} \bigvee_{\chi(\vec{z}) \in \mathcal{P}, t^A(\vec{z})} (\exists \vec{z}) (\chi(\vec{z}) \wedge x = t(\vec{z})) \right),$$

where the disjunction is taken over all the formulae $\chi(\vec{z})$ in $\mathcal{P}$ and all the terms $t^A(\vec{z})$;

- (v) for any sort A of Σ , any formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$, where $\vec{x} = (x_1^{A_1}, \dots, x_n^{A_n})$ and $\vec{y} = (y_1^{B_1}, \dots, y_m^{B_m})$, and any terms $t^A(\vec{x})$ and $s^A(\vec{y})$, the sequent

$$\left(\phi(\vec{x}) \wedge \psi(\vec{y}) \wedge t(\vec{x}) = s(\vec{y}) \vdash_{\vec{x}, \vec{y}} \bigvee_{\substack{\chi(\vec{z}) \in \mathcal{P}, p_1^{A_1}(\vec{z}), \dots, p_n^{A_n}(\vec{z}) \\ q_1^{B_1}(\vec{z}), \dots, q_m^{B_m}(\vec{z})}} (\exists \vec{z}) \left(\chi(\vec{z}) \wedge \bigwedge_{\substack{i \in \{1, \dots, n\}, \\ j \in \{1, \dots, m\}}} (x_i = p_i(\vec{z}) \wedge y_j = q_j(\vec{z})) \right) \right),$$

where the disjunction is taken over all the formulae $\chi(\vec{z})$ in $\mathcal{P}$ and all the tuples of terms $p_1^{A_1}(\vec{z}), \dots, p_n^{A_n}(\vec{z})$ and $q_1^{B_1}(\vec{z}), \dots, q_m^{B_m}(\vec{z})$ such that

$$(p_1(\vec{\xi}_\chi), \dots, p_n(\vec{\xi}_\chi)) \in [[\vec{x} \cdot \phi]]_{M_{\{\vec{z} \cdot \chi\}}},$$

$$(q_1(\vec{\xi}_\chi), \dots, q_m(\vec{\xi}_\chi)) \in [[\vec{y} \cdot \psi]]_{M_{\{\vec{z} \cdot \chi\}}}$$

and

$$t(p_1(\vec{\xi}_\chi), \dots, p_n(\vec{\xi}_\chi)) = s(q_1(\vec{\xi}_\chi), \dots, q_m(\vec{\xi}_\chi)) \text{ in } M_{\{\vec{z} \cdot \chi\}}.$$

If moreover $\mathbb{T}$ is of presheaf type (in which case every finitely presentable $\mathbb{T}$ -model is strongly finitely presentable – cf. Corollary 6.2.8) then the quotient axiomatized by these sequents coincides with the theory $\mathbb{T}_\mathcal{K}$ defined in Theorem 8.2.10.

Proof Let $\mathbb{R}$ be the geometric theory obtained from $\mathbb{T}$ by adding all the sequents specified above. The objects of $\mathcal{K}$ are clearly models of $\mathbb{R}$ (cf. Remarks 6.3.7(a) and 6.3.11); since they are strongly finitely presentable as $\mathbb{T}$ -models, it follows from Proposition 6.3.12(ii) that condition (iii) of Theorem 6.3.1 is satisfied by the theory $\mathbb{R}$ with respect to the category $\mathcal{K}$. By Remarks 6.3.7(a) and 6.3.11, $\mathbb{R}$ also satisfies conditions (i) and (ii) of Theorem 6.3.1 with respect to $\mathcal{K}$. Theorem 6.3.1 thus implies that $\mathbb{R}$ is of presheaf type classified by the topos $[\mathcal{K}, \mathbf{Set}]$ via a Morita equivalence induced by the canonical geometric morphism $p_\mathcal{K} : [\mathcal{K}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}_\mathbb{T}, J_\mathbb{T})$; this ensures in particular that if $\mathbb{T}$ is of presheaf type then $\mathbb{R}$ is equal to the theory $\mathbb{T}_\mathcal{K}$. $\square$

Remark 8.2.14 Theorem 8.2.13 has been applied in [33] to describe geometric theories classified by Connes' cyclic topos ([35]), Connes-Consani's epicyclic topos ([37]) and Connes-Consani's arithmetic topos ([36]). It has also been used in [32] to obtain a simple axiomatization for the geometric theory of finite MV-chains (cf. Theorem 9.7 therein).

